START HERE · OPEN THIS PAGE FIRST

Your placement kit

Three PDFs and one live site. Everything on this page is tappable: rows jump to the right page, underlined topic names anywhere open their in-depth page online, and every section has an ↑ **Index** button to come back. Use the bookmarks panel in your PDF viewer for the full outline.

Your three PDFs

This PDF · Complete
Guide + Cheat Sheet + Roadmap. Open it to plan, look things up and check progress.

Knowledge Base PDF
97 topics, one per page, works offline. Online version: Placement Knowledge Base.

Playbook PDF
The Café-theme Playbook: Today, all 13 parts. Tick progress in the online playbook; the PDF is a snapshot.

Inside this PDF

1 · Placement Field Guide p. 3
Levels, efficiency system, every ladder, off-campus machine, stories

2 · Prep Cheat Sheet p. 23
Rules, complexity, STL, patterns, templates, DP, Core CS, SQL, security, AWS

3 · Placement Roadmap p. 32
Timeline, phases, week-by-week and day-by-day plan, milestones

Every day in five steps

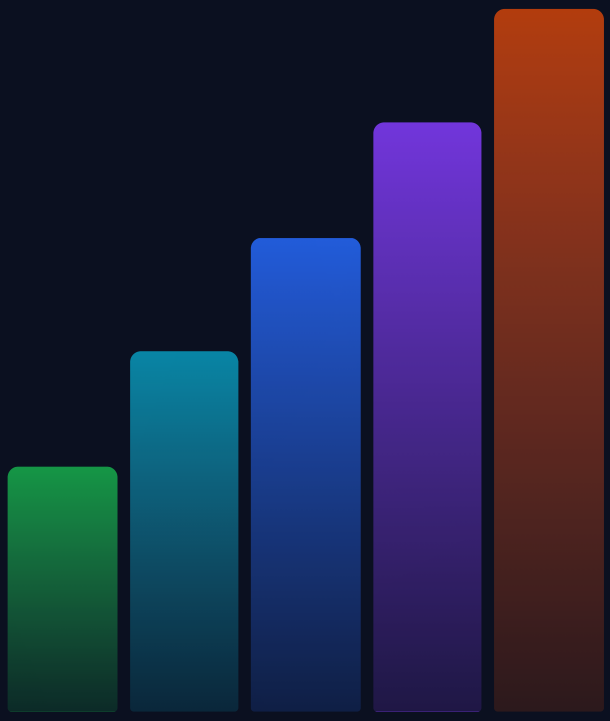
1. **Pick your day mode** (full force, college, exam, low energy, sprint).
2. **Revise first:** problems due on your 1-3-7-21 sheet.
3. **New work in 50/10 blocks:** Striver in order, then LeetCode, Core CS, cyber, AWS.
4. **Stuck or unsure?** Tap the topic name → Knowledge Base page.
5. **Night, 5 min:** error log + tomorrow's 3 problems. **Sunday:** weekly review.
6. **Never zero:** one problem touched keeps the streak.

Where to find what

WHEN	I WANT TO…	PAGE
TODAY	**Plan your day** · Playbook PDF → Today, or the Day modes	p. 7 →
TODAY	**Stuck on a problem** · The 7-step routine and the 20–40 rule	p. 8 →
TODAY	**Which pattern is this?** · 20 triggers and the constraint table	p. 12 →
TODAY	**Copy a C++ template** · Binary search, sliding window, graphs, DSU…	p. 26 →
TODAY	**Solve a DP problem** · The 5-step DP recipe	p. 27 →
THIS WEEK	**This week's plan** · Every week and month of the plan	p. 35 →
THIS WEEK	**Learn faster** · 12 methods + AI prompts + 80/20 list	p. 6 →
THIS WEEK	**Core CS questions** · OS, DBMS, CN, OOP, SQL by level	p. 13 →
THIS WEEK	**Security ladder** · PortSwigger → pwn.college → bug bounty	p. 15 →
THIS MONTH	**Am I on track?** · Milestone checklists by date	p. 21 →
THIS MONTH	**Roadmap milestones** · Targets and dates	p. 74 →
APPLYING	**Apply off-campus** · Channels, quotas, referral + cold email templates	p. 17 →
APPLYING	**Interview coming up** · Rounds, STAR stories, offers, mock schedule	p. 18 →
MOTIVATION	**Tier-3 stories** · 16 students who did it, with sources	p. 19 →

Export your playbook progress on 18 Nov (Logs → Back up progress) before Claude Pro ends on 19 Nov.

FIELD GUIDE · NOV 2026 → JUN 2028



Tier-3 to Top Offer

The placement field guide for a B.Tech CSE (Cybersecurity) student: DSA, Core CS, system design, security, AWS and AI, sorted into five difficulty levels, with a complete efficiency system (day modes, a 7-step problem routine, revision schedule, AI prompts, OA tactics), an off-campus hiring machine, and 16 real stories from students who did it from tier-3 colleges.

L0 · Foundation | L1 · Core | L2 · Interview-ready | L3 · Advanced | L4 · Elite

BUILT FROM
your Playbook, Roadmap and Cheat Sheet, plus public interview experiences, free courses and hiring pages

TAP ANY TOPIC
Underlined names open an in-depth page in the Placement Knowledge Base: concept, why it matters for you, code, mistakes, interview questions, practice

TRACK
Intensive: Striver core done by 30 Jan 2027 · AWS SAA by April · sprint 31 May–31 Jul · placements Aug–Dec 2027

Index

Placement Field Guide. Tap any line to jump; every page has ↑ Index to come back.

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START HERE

00 How this guide works

↑ Index

Every topic is a ladder. Find your rung, climb one at a time, and stop climbing where the target job stops asking.

L0
Foundation
You can't do it yet. Learn the syntax, the idea, the vocabulary. Goal: no blank page.

L1
Core
You can do it with notes. Standard problems, standard questions. Service-company OA level.

L2
Interview-ready
You can do it cold, timed, out loud. Product-company interview level (Amazon, Razorpay, Atlassian).

L3
Advanced
You can do the twist. Hards, contests, design trade-offs. Google, Microsoft, D. E. Shaw level.

L4
Elite
You do what few freshers do. CP rating 1900+, CVEs, open source, HFT puzzles. The outlier offers.

Which level each target needs

TARGET	DSA	CORE CS	DESIGN	SECURITY	CLOUD/AI
Mass recruiters (TCS Digital, Infosys SP, Accenture)	L1 · Core	L1 · Core	—	—	—
Product mid-tier (Razorpay, Zomato, Agoda, Myntra)	L2 · Interview-ready	L2 · Interview-ready	L1 · Core LLD	—	L1 · Core
Big Tech (Amazon, Microsoft, Google, Atlassian)	L3 · Advanced	L2 · Interview-ready	L2 · Interview-ready	—	L1 · Core
Security roles (AppSec, cloud security, SOC+)	L2 · Interview-ready	L2 · Interview-ready	L1 · Core	L3 · Advanced	L2 · Interview-ready AWS
HFT / quant dev (Tower, Graviton, Quadeye)	L4 · Elite	L3 · Advanced	L2 · Interview-ready	—	maths L3 · Advanced

EVERY NAME IS A DOOR
Topic names with a dotted underline link to their own page in the Placement Knowledge Base: the concept in depth, why it matters for your plan, key points, code, common mistakes, interview questions to tick off, practice problems and free resources. 97 topics.

THE ONE RULE
Depth on one ladder beats breadth on five. Your main ladder is DSA to L3 · Advanced. Core CS to L2 · Interview-ready. Security to L3 · Advanced because it's your degree and your edge. Everything else stops at L1 · Core – L2 · Interview-ready unless a specific job needs more.

01 THE ROUTE
Nov 2026 → Jun 2028 on one page ↑ Index

The intensive track you chose, with the semester and exams built in.

16 Nov – 13 Dec 2026 **Full force**
C++ in 2 days, then Striver sheet 5–6/day in order, LeetCode daily, 1 Core CS subject, cyber 1 h, AWS 30 min

14 Dec – 30 Jan 2027 **Sheet core done**
College 1–2 days/week. Finish Striver core (30 Jan). AI track starts. TenantWall build: CloudTrail, metric filter, alarm, SNS

Feb – 21 Mar 2027 **Hard pass 1 + paper**
Hard problems part 1, CP31 + Codeforces Div 2/3, OS + DBMS + CN revision, TenantWall paper by 21 Mar

22 Mar – 7 Apr 2027 **Semester exams**
Exams first. DSA drops to 1 revision problem/day. Nothing new.

8 Apr – 30 May 2027 **Hard pass 2 + AWS**
SAA-C03 exam week of 19 Apr. Hard part 2, LLD, 2 big projects, binary-exploitation taster

31 May – 31 Jul 2027 **Placement sprint**
2 mocks/week, company-tagged LeetCode, OA simulations, resume v5, 25 applications/week off-campus

Aug – Dec 2027 **Placements**
On-campus drives + off-campus machine running in parallel. Keep 3 problems/day.

Jan – Jun 2028 **Convert / upgrade**
Internship-to-PPO or keep interviewing for a better offer until graduation

Internship windows you can still hit

WINDOW	WHAT OPENS	ACTION NOW
Nov 2026 – Feb 2027	Off-campus summer 2027 intern roles still open at startups and mid-tier product cos; Unstop competitions (Flipkart GRiD, Myntra HackerRamp, etc. run yearly)	Apply weekly from mid-Dec once 150+ problems are done
Feb – Apr 2027	GSoC contributor applications (usually Mar–Apr); LFX Mentorship spring cohort	Pick 1 org in Jan, make 3 merged PRs before applying
Jun – Sep 2027	Big Tech new-grad + 6-month intern roles for the 2028 batch	Have resume, 2 projects, 400 problems ready by 31 May
Year-round	Bug bounty programmes, CTFs (KPMG, CyberSecurity events), open-source issues	One CTF a month from Feb

NOC NOTE
Your college gives no NOC by default, but you said a good company internship changes that. Have the offer letter first, then ask. Remote internships and 6-month Jan–Jun 2028 roles avoid most attendance clashes.

LEARN FASTER

02 The efficiency system

↑ Index

Twelve methods with evidence behind them. Use all of them every week; they cost minutes and save months.

01
Retrieval first

Close the video and solve or explain from memory before checking. Testing yourself beats rereading for long-term memory (Roediger & Karpicke, 2006).

02
Spaced revision 1-3-7-21

Re-solve each problem you got wrong after 1, 3, 7 and 21 days. Keep the list in a sheet; it's your most valuable file.

03
The 20-40 rule

Stuck 20 min on Easy/Medium (40 on Hard)? Read only the hint or the approach line, then code it yourself. Never copy code.

04
Pattern, not problem

After each solve, write one line: 'trigger → pattern'. E.g. 'sorted + pair sum → two pointers'. 300 problems become ~25 patterns.

05
Interleave

Mix topics in revision days (a graph, a DP, a string). Blocked practice feels faster; interleaved practice sticks better.

06
Speak while you code

Solve 1 problem a day out loud as if in an interview. It's a separate skill and the one tier-3 candidates practise least.

07
Error log

Every wrong submission: the bug in one line (off-by-one, overflow, missed empty case). Read the log before each contest.

08
Upsolve

After every contest, solve the next unsolved problem with the editorial. That one problem teaches more than the whole contest.

09
Feynman for Core CS

Explain deadlock, normalisation or TCP handshake on a blank page in 5 lines. Can't? That's your revision list.

10
50/10 deep blocks

50 min phone-in-another-room, 10 min break. 6 blocks/day is a full-force day. Track blocks, not hours.

11
Sleep 7+ h

Memory consolidates during sleep; all-nighters before contests cost more than they give.

12
Weekly review (Sun, 30 min)

What did I solve, what did I fail, what moves next week? Adjust the plan instead of abandoning it.

A full-force day (16 Nov → 13 Dec)

BLOCK	TIME	WORK
1–2	09:30–11:30	Striver sheet: 3 new problems in order (C++)
3	11:45–12:35	LeetCode: 1 daily + 1 revision from the 1-3-7-21 list
4	14:00–14:50	Core CS: one subtopic, Feynman page, 10 MCQs
5	15:00–15:50	Striver: 2–3 more problems
6	17:00–18:00	Cyber: one PortSwigger / TryHackMe lab, timed
7	18:15–18:45	AWS 30 min (this week's module)
8	21:00–21:30	Speak-aloud problem + error log + tomorrow's 3 problems picked

02·A EFFICIENT PROCESS

Run your prep like a system

↑ Index

Decide once, not every morning. Pick today's mode, follow its recipe, and let the loops catch what slips.

Five day modes: pick one each morning in 10 seconds

MODE	WHEN	MINIMUM THAT COUNTS AS A WIN	DROP
Full force	Home, no college	6–8 deep blocks: 5–6 Striver, 1 LC, Core CS, cyber, AWS	nothing
College day	Attending (1–2/week)	2 problems on the commute/evening + AWS 30 min + 10 Core CS flashcards	cyber, new topics
Exam mode	22 Mar – 7 Apr	1 revision problem from the 1-3-7-21 list	everything else
Low energy	Tired, ill, bad day	1 Easy re-solve + read your error log (20 min)	the guilt
Sprint	31 May – 31 Jul	Mock or OA sim + company-tagged set + 5 applications	new learning

> **NEVER-ZERO RULE**
> The streak is "at least one problem touched", not "a perfect day". A low-energy day that keeps the streak is worth more than a perfect day followed by four days off.

Four loops that keep the plan honest

Daily · 5 min at night
Pick tomorrow's 3 problems and 1 Core CS topic. Write them down. Mornings start with no decisions.

Weekly · Sun 30 min
Count problems, contests, blocks. Re-solve the week's failed problems. Move unfinished items, never stack them.

Monthly · last Sun 1 h
Check the milestone box (section 13). One timed mock on old topics. Cut whatever isn't moving a number.

Phase · each milestone
Rewrite the resume, update LinkedIn/GitHub, adjust the next phase's daily quota up or down by one problem.

Decision rules: so you never waste an hour deciding

- **Stuck?** 20 min (Easy/Med) or 40 min (Hard) → read only the approach → code it yourself → add to the 1-3-7-21 list.
- **Video or article?** Article first (faster, searchable). Video only if the article didn't click, at 1.5×.
- **New resource tempting you?** Only if the current one has no material on that topic. Finish the sheet before switching.
- **Behind schedule?** Cut the newest track (AI, then HFT), never DSA or revision. Never double tomorrow's quota.
- **Two good options?** Pick the one that produces a proof (solved problem, lab, PR, resume line).
- **Feeling ready to apply?** You never will. Apply when the checklist says so (≈150 problems + 1 project).

EFFICIENT PROCESS

02·B One problem, done the efficient way

↑ Index

The same 7 steps for every problem. This turns 550 problems into skill instead of a number.

1
Read + restate (2 min)
Rewrite the task in one line. Note n's limit → target complexity (table in section 04). List 3 edge cases: empty, one element, max/duplicates/negatives.

2
Brute force first (3 min)
Say or write the simplest correct idea and its complexity. Interviewers reward this; it also gives you a checker.

3
Name the pattern (2 min)
Use the trigger table. If none fits, try: sort it? hash it? binary search the answer? think in states (DP)?

4
Code it (≤15 min)
Clean names, no premature tricks. C++ template below. Stop at the timebox and take the hint.

5
Dry run (3 min)
Trace one normal + one edge case by hand before submitting. Catches most WA.

6
Log it (1 min)
Sheet row: link · pattern · trigger line · mistake · next revision date. This is the most valuable minute.

7
Revise on schedule
Day 1, 3, 7, 21: re-solve from a blank file in ≤10 min. Pass twice in a row → retire it.

Tracker sheet columns (Google Sheets, free)

`Date · Problem · Link · Level · Pattern · Trigger · Time · Hint? · Mistake · R1 · R3 · R7 · R21`

Formula for next revision: `=A2+1`, `=A2+3`... then filter "due ≤ today" each morning. 10 minutes of setup saves hundreds of re-solves.

Reading an editorial efficiently

1. Read only until you see the key idea, then close it.
2. Code it yourself without looking.
3. Only then read their code and compare.
4. Write the one line you missed.

Contest C++ template (compile-checked)

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;
#define all(x) (x).begin(), (x).end()
const ll INF = 4e18; const int MOD = 1e9 + 7;
void solve() {
    int n; cin >> n;
    vector<ll> a(n);
    for (auto &x : a) cin >> x;
    // brute force first, then optimise
}
int main() {
    ios::sync_with_stdio(false); cin.tie(nullptr);
    int t = 1; cin >> t;
    while (t--) solve();
}
```

Online assessment and contest tactics

Read all problems first
2 minutes. Solve in order of confidence, not order given.

Partial scores count
HackerRank-style OAs score test cases. A correct brute force on half the tests beats a perfect idea not submitted.

Watch overflow
Sums/products of ints → `long long`. Most silent WA in OAs.

Custom tests
Run the edge cases you listed before submitting. Empty, 1, max n, all equal.

Last 10 minutes
Stop optimising. Submit what passes, add guards for edge cases.

Same environment
Practise on HackerRank/HackerEarth too; their I/O and editors differ from LeetCode.

02·C EFFICIENT PROCESS
Use AI, tools and the 80/20 rule ↑ Index

AI is a tutor and interviewer, never your hands. Tools remove friction. The 80/20 list tells you what to learn first.

Six prompts for Claude or Gemini

HINT-ONLY TUTOR
I'm solving [problem link/text]. Don't give the solution or code. Give me one hint at a time, the smallest nudge first. Wait for me to say 'next' before the next hint.

MOCK INTERVIEWER
Act as an Amazon SDE interviewer. Give me one Medium DSA problem. Stay silent while I think aloud, ask about edge cases and complexity, then score me 1–5 on problem solving, code and communication with one fix each.

CODE REVIEWER
Here is my accepted C++ solution. Don't rewrite it. Point out: bugs on edge cases, complexity, unclear names, and one cleaner STL idiom I could have used.

FEYNMAN CHECKER
I'll explain [deadlock / B+ tree / TCP handshake] in 5 lines. Find what's wrong or missing, then ask me 3 interview follow-up questions on it.

FLASHCARD MAKER
Turn these notes into 10 question–answer flashcards, one fact each, in 'Q; A' format for Anki import.

RESUME REVIEWER
Review this resume for an SDE intern role at a product company. For each bullet: is there a number? an action verb? a technology? Rewrite the weakest 3.

THE AI RULE
Never paste a problem and ask for the solution before your 20/40-minute timebox is up. Copying AI code feels like progress and gives zero interview skill — interviewers can tell in 2 minutes.

The 20% that gets 80% of the questions

AREA	LEARN THESE FIRST
DSA	Arrays + hashing, two pointers, sliding window, binary search, trees (traversals, BST, LCA), graphs (BFS/DFS, topo, Dijkstra), DP (1D, 2D grid, knapsack, LIS, LCS), heaps, stacks
OS	Process/thread, scheduling, deadlock, paging + virtual memory, synchronisation (mutex, semaphore)
DBMS + SQL	Normalisation, ACID + isolation, indexing, joins, GROUP BY + HAVING, window functions, Nth highest
CN	OSI/TCP-IP, TCP vs UDP, handshake, HTTP/HTTPS/TLS, DNS, 'type a URL' walkthrough
OOP	4 pillars, virtual/vtable, overloading vs overriding, SOLID, 3 patterns
Security	OWASP Top 10, auth vs authz, hashing vs encryption, TLS, IAM least privilege

Free tool stack

NEED	TOOL	WHY IT SAVES TIME
Tracker + revision dates	Google Sheets	Filter 'due today' in one click
Core CS memory	Anki	Spaced repetition done for you, 10 min/day
Focus blocks	Pomofocus + phone in another room	Counts blocks, not hours
Diagrams (LLD/HLD, TenantWall)	Excalidraw, diagrams.net (AWS icons)	Fast, exportable, looks pro
Contest calendar	clist.by	All contests in one place
Mock interviews	Pramp, peers, AI mock prompt	Free, real-time speaking practice
Job alerts	LinkedIn + Naukri + company career pages saved searches	Jobs come to you; apply within 48 h

02·D EFFICIENT PROCESS

Apply, interview and last the distance

↑ Index

Batch the boring parts, reuse everything, and protect the energy that does the real work.

Application pipeline in 90 min/week

1. **Master resume** with every bullet you have (2 pages, private).
2. **3 role versions** cut from it: SDE, Security/AppSec, Cloud. One page each.
3. **Tailor in 5 min**: swap 2 bullets and the skills order to match the job's top 5 keywords. Nothing more.
4. **Batch**: Mon + Thu evening, 45 min each. Alerts collected, applications sent, referral messages sent, tracker updated.
5. **Follow up** once after 5–7 days, from the tracker's "next action" column.
6. **Reuse answers**: keep a doc of "Why this company / Tell me about yourself / Strengths" answers, edit per company.

Company-specific prep in 1 week

1. Read 5–10 recent interview experiences for that company and role (LeetCode Discuss, GfG).
2. Solve its last-6-months LeetCode tag list, top 30 by frequency.
3. Map its values (e.g. Amazon Leadership Principles) to your 6 STAR stories.
4. Know its product: use it, find one thing you'd improve, one security thought.

Interview day: efficient habits

- Night before: no new topics. Re-read your error log, pattern sheet and STAR stories.
- Have a 60-second "tell me about yourself": who you are, 2 proofs (rating, TenantWall), why this role.
- Think aloud from second one; silence is scored as stuck.
- State brute force, then improve. Ask before coding: "Shall I code this approach?"
- End every round with one smart question about the team's work.
- After: write down every question within 1 hour. That list is your next week's revision.

Energy: the hidden multiplier

- **Sleep 7–8 h** at a fixed time; it's when memory consolidates.
- **Hardest work first** (new DSA) in your sharpest hours; revision and AWS when tired.
- **Move 20–30 min** daily (walk, cricket, gym). Improves focus more than coffee.
- **Phone out of the room** during blocks; social apps only after block 4.
- **One rest half-day a week.** Burnout costs weeks; rest costs hours.
- **Compare with your last month**, not with LinkedIn posts.

> **EFFICIENCY IN ONE LINE**
> Fixed modes, a 7-step problem routine, a revision sheet that tells you what to redo, AI as a tutor not a solver, batched applications, and sleep. Everything else in this guide plugs into those six.

03 MAIN LADDER

DSA from L0 to L4 ↑ Index

The ladder that decides most offers. Climb it in order; don't skip L2 to chase L4.

L0 FOUNDATION

C++ and STL in 2 days

vector, string, pair, map, unordered_map, set, priority_queue, stack, queue, deque; sort with lambdas; iterators; `auto`; fast I/O. Big-O of every operation.

Use: takeuforward STL, your Cheat Sheet §3, learncpp.com (chapters you need)

Done when: you can write any STL container op without searching

L1 CORE

Striver A2Z: steps 1–9

Basics, sorting, arrays, binary search, strings, linked list, recursion, bit manipulation, stack/queue. Easy + Medium in sheet order, 5–6 a day.

Use: Striver A2Z sheet, LeetCode daily

Done when: you solve 80% of step problems within 25 min without hints

L2 INTERVIEW-READY

Striver steps 10–18 + patterns

Sliding window, two pointers, heaps, greedy, binary trees, BST, graphs (BFS/DFS, Dijkstra, topo sort, DSU), DP 1D/2D/subsequence/LIS/strings, tries.

Use: Striver sheet, NeetCode 150 as a cross-check, LeetCode company tags

Done when: LC Medium in ≤25 min, out loud, with complexity — 70% of the time

L3 ADVANCED

Hard problems + contests

Hards from the sheet in 2 passes; monotonic stack, interval DP, bitmask DP, DP on trees, segment tree / BIT, string hashing, KMP/Z. Weekly LC contest + Codeforces Div 2.

Use: CSES, CP31 sheet, Codeforces, cp-algorithms

Done when: LC contest rating 1800+ or CF 1400+; LC Hard in 45 min half the time

L4 ELITE

Competitive edge

CF Div 1/2 D–E, advanced graphs (flows, SCC, bridges), heavy-light, centroid, matrix exponentiation, probability DP, game theory. ICPC-style team practice.

Use: CSES advanced sections, USACO Guide gold/platinum, CF gym, AtCoder ABC/ARC

Done when: CF 1900+ / LC Knight-Guardian; needed only for HFT/quant-dev or top-tier Google

Problem counts that are enough

BY	TOTAL SOLVED	MIX	SIGNAL
30 Jan 2027	≈ 300	Striver core E/M	L1 · Core → L2 · Interview-ready
30 May 2027	≈ 450	+ 120 Hards/tagged	L2 · Interview-ready → L3 · Advanced
31 Jul 2027	≈ 550	+ company-tagged, mocks	L3 · Advanced

04 PATTERN TRIGGERS
Read the problem, name the pattern ↑ Index

20 triggers that cover most interview problems. Cover the right column and quiz yourself.

IF THE PROBLEM SAYS…	REACH FOR	LEVEL
Sorted array, find pair/triple	**Two pointers**	L1 · Core
Longest/shortest subarray with condition	**Sliding window**	L1 · Core
Min/max answer that is monotone	**Binary search on answer**	L2 · Interview-ready
Next greater / smaller	**Monotonic stack**	L2 · Interview-ready
k largest / streaming median	**Heap(s)**	L2 · Interview-ready
Subarray sum = k	**Prefix sum + hashmap**	L1 · Core
All subsets / permutations	**Backtracking**	L1 · Core
Shortest path, unweighted	**BFS**	L1 · Core
Shortest path, weighted ≥0	**Dijkstra**	L2 · Interview-ready
Dependencies / order	**Topological sort**	L2 · Interview-ready
Connected groups, merging	**DSU**	L2 · Interview-ready
Count ways / min cost over choices	**DP**	L2 · Interview-ready
Two strings compare	**2D DP (LCS/edit)**	L2 · Interview-ready
Prefix matching many words	**Trie**	L2 · Interview-ready
Range query + update	**Segment tree / BIT**	L3 · Advanced
Intervals overlap	**Sort + sweep**	L2 · Interview-ready
n ≤ 20, choose subsets	**Bitmask DP**	L3 · Advanced
Tree answer per node	**DFS + DP on trees**	L3 · Advanced
Max XOR pair	**Bit trie**	L3 · Advanced
Pattern in text, O(n)	**KMP / Z / hashing**	L3 · Advanced

Constraint → complexity

N UP TO	10	20	500	5 000	10^6	10^9+
Aim for	O(n!)	$O(2^n)$	$O(n^3)$	$O(n^2)$	O(n log n)	O(log n) / O(1)

THE INTERVIEW LOOP (UMPIRE)

Understand (repeat it, ask about constraints and edge cases) · **M**atch a pattern · **P**lan in words, state complexity · **I**mplement cleanly · **R**eview with a dry run · **E**valuate trade-offs and a better approach. Talk the whole time.

05 CORE CS
OS, DBMS, CN, OOP, SQL by level ↑ Index

Interviewers ask the same 40 questions. Know them at L2 and you've covered most rounds.

SUBJECT	L1 · Core KNOW	L2 · Interview-ready EXPLAIN + EXAMPLE	L3 · Advanced DEPTH
OS	process vs thread, scheduling algos, deadlock conditions	paging, virtual memory, page faults, mutex vs semaphore, producer-consumer, thrashing	TLB, copy-on-write, context-switch cost, Linux scheduler (CFS), memory-mapped files
DBMS	keys, ER, normal forms to 3NF, ACID	BCNF, indexing (B+ tree), transactions and isolation levels, deadlock, joins	MVCC, write-ahead logging, query plans, sharding vs replication, CAP
CN	OSI vs TCP/IP, TCP vs UDP, IP classes, DNS	3-way handshake, flow vs congestion control, HTTP/HTTPS + TLS, subnetting, what happens when you type a URL	TCP congestion algorithms, QUIC/HTTP3, BGP basics, NAT traversal, CDN
OOP	4 pillars with C++ examples	virtual functions + vtable, overloading vs overriding, abstract vs interface, SOLID	design patterns (singleton, factory, observer, strategy), RAII, rule of 5
SQL	SELECT, WHERE, GROUP BY, HAVING, JOIN	subqueries, window functions (RANK, ROW_NUMBER, LAG), Nth highest salary	query optimisation, indexes in practice, CTE recursion

Use: GfG OS · Striver/takeuforward Core CS notes · Gate Smashers for weak topics · LeetCode SQL 50 · SQLBolt

The 12 most-asked questions (answer each in 60 seconds)

1. Process vs thread — and why threads are cheaper
2. Four conditions of deadlock and how to prevent one
3. What is virtual memory and why do we need it?
4. Mutex vs semaphore with an example
5. Explain normalisation up to 3NF using one table
6. What are ACID properties? Which isolation level prevents dirty reads?
7. How does a B+ tree index speed up a query?
8. What happens when you type google.com and press Enter?
9. TCP vs UDP — when would you choose UDP?
10. Explain the TCP 3-way handshake and why not 2-way
11. Runtime polymorphism in C++: how does the vtable work?
12. Write SQL for the second-highest salary per department

06 BUILD AND DESIGN
System design and projects ↑ Index

Freshers get LLD more than HLD. Projects get you shortlisted; design talk gets you hired.

L1 CORE

LLD basics

Classes, relationships, SOLID, 4 patterns. Design a parking lot, library system, Splitwise, elevator in C++.

Use: awesome-low-level-design (free)

Done when: you can design a parking lot with classes in 30 min

L2 INTERVIEW-READY

HLD fundamentals

Load balancer, caching, CDN, DB choice (SQL vs NoSQL), replication, sharding, message queues, rate limiting. Design URL shortener, chat app, news feed.

Use: system-design-primer, roadmap.sh, your AWS SAA notes

Done when: you can draw and defend a URL shortener with numbers

L3 ADVANCED

Trade-off depth

Consistency models, CAP/PACELC, idempotency, distributed IDs, back-pressure, observability.

Use: Designing Data-Intensive Applications (Kleppmann) chapters 1–9; engineering blogs (Razorpay, Zomato, Uber)

Done when: you can argue two designs and pick one with reasons

Projects that work for a cyber + SDE profile

PROJECT	SHOWS	LEVEL
TenantWall: ABAC multi-tenant isolation on AWS with CloudTrail, metric-filter alarm, SNS, Access Analyzer + paper	cloud security, IAM depth, research	L3 · Advanced
Rate-limited URL shortener (C++/Go or Node) with Redis cache, load test numbers, deployed on AWS	backend + system design	L2 · Interview-ready
Log-based intrusion detector: parse auth/nginx logs, flag brute force and scans, dashboard	security + data handling	L2 · Interview-ready
Secure AI assistant: RAG over your notes with prompt-injection tests and guardrails	AI + AppSec (rare combo)	L3 · Advanced
Open-source PRs (3+ merged) in one org	real code review, collaboration	L3 · Advanced

PROJECT RULE

Every project needs: a live link or demo video, a README with an architecture diagram, one number (latency, users, requests/s, bugs found), and a line on what you'd do next. Two great projects beat six tutorials.

07 YOUR EDGE
Security ladder ↑ Index

Your degree is cybersecurity. Most SDE candidates can't talk about it — use that in every interview.

L0 FOUNDATION

Foundations

Linux command line, networking basics, HTTP, how the web works.

Use: OverTheWire Bandit, TryHackMe Pre-Security (free rooms)

Done when: Bandit level 20+

L1 CORE

Web security

OWASP Top 10: SQLi, XSS, CSRF, SSRF, IDOR, auth flaws, access control.

Use: PortSwigger Web Security Academy (free)

Done when: all Apprentice labs + 30 Practitioner labs

L2 INTERVIEW-READY

Practical AppSec + cloud

Burp Suite workflow, code review for vulns, secure coding in C++, AWS IAM/S3 misconfigurations, threat modelling (STRIDE).

Use: PortSwigger Practitioner, flaws.cloud, OWASP

Done when: you can explain and fix 5 vulns in code

L3 ADVANCED

Binary + CTFs

C memory, stack layout, buffer overflow, format strings, ROP basics, GDB/pwndbg, Ghidra reversing. Jeopardy CTFs monthly.

Use: pwn.college, picoCTF, CTFtime

Done when: top-25% finish in a mid CTF; 3 write-ups on a blog

L4 ELITE

Proof in the wild

Valid bug bounty reports, a CVE, a security tool on GitHub with users, conference/college talk.

Use: HackerOne, Bugcrowd, vendor disclosure programmes

Done when: one accepted report or CVE

USE IT IN SDE INTERVIEWS

When asked about any project: "I also threat-modelled it — here's the IDOR I found and fixed." When asked about databases: mention SQL injection and parameterised queries. One security sentence per answer makes you memorable.

08 MULTIPLIERS

AWS, AI and the HFT stretch ↑ Index

Each one adds a line to your resume that most tier-3 candidates don't have.

TRACK	L1 · Core	L2 · Interview-ready	L3 · Advanced
AWS (30 min daily)	IAM, EC2, S3, VPC basics; billing alarm	AWS Academy Cloud Architecting done; SAA-C03 passed (Apr 2027)	TenantWall in production-style; security specialty later
AI (≤1 h from 14 Dec)	Python, NumPy, how LLMs work, prompting	Build RAG + tool-calling app; evals	Prompt-injection testing, AI security (OWASP LLM Top 10)
HFT maths (Apr–Jul 2027)	Counting, Bayes, expectation	Green-book probability, Brainstellar puzzles, Zetamac 40+	Market-making games, CF 1600+, low-latency C++

Use: AWS Academy (via your faculty) · freeCodeCamp SAA course · DeepLearning.AI short courses · OWASP LLM Top 10 · Stat 110 · Brainstellar · Zetamac

Resume: one page, ATS-safe

Order

1. Name, phone, email, LinkedIn, GitHub, LeetCode/CF links
2. Education (CGPA if ≥ 7.5)
3. Experience / internships
4. Projects (2–3, each 2–3 bullets)
5. Achievements (ratings, CTF ranks, SAA cert, paper)
6. Skills (languages, tools, cloud)

Bullet formula

`Did X using Y, which resulted in Z (number)`

Weak: Made a URL shortener using Node.

Strong: Built a URL shortener serving 1.2k req/s on one t3.micro by adding Redis caching and a token-bucket rate limiter; p95 latency 18 ms.

Single column, no photos, no tables, PDF, file name `Firstname_Lastname_SDE.pdf`.

09 OFF-CAMPUS

The hiring machine

↑ Index

On-campus is a lottery you can't control. Off-campus is a numbers game you can.

CHANNEL	HOW	WEEKLY QUOTA
Referrals	Find alumni/employees on LinkedIn at target companies; send the referral note below with job ID + resume	10 messages
Careers pages	Bookmark 40 companies' careers pages; check Mon + Thu. Apply within 48 h of posting	10 applications
Hiring contests	Unstop, HackerEarth, Naukri Campus, company hackathons (Flipkart GRiD, Myntra HackerRamp, Amazon ML Summer School, JPMC Code for Good)	every one that opens
Cold email	Startup founders/CTOs (YC India, funded on Inc42/Entrackr) with a 5-line email + one relevant project	5 emails
Open source	GSoC, LFX Mentorship, MLH Fellowship; maintainers refer contributors	2 PRs
Contest ratings	Recruiters filter on LC/CF rating; some firms invite top contest rankers	1–2 contests

Referral message (LinkedIn, 300 chars)

Hi [Name], I'm a 3rd-year CSE (Cybersecurity) student who has solved 450+ DSA problems and built [project — one line]. I'm applying for [Role, Job ID] at [Company]. Would you be open to referring me? Resume: [link]. Happy to share anything else. Thank you!

Cold email to a startup

Subject: SDE intern — built [X] relevant to [their product]

Hi [Name], I use/studied [product]. I'm a CSE (Cyber) student; I built [project + number], and I noticed [specific thing — a bug, a security header missing, a feature idea]. I'd love to intern with your team for [months]. Resume + GitHub: [links]. Could we talk for 15 minutes this week?

TRACK IT

One sheet: company, role, date, channel, contact, status, next follow-up. Follow up once after 5–7 days. Expect ~2–5% replies; 50 good applications a month is what produces interviews.

10 THE ROOM

Interviews, rounds and offers

↑ Index

What each round checks, and how to prepare for it.

ROUND	WHAT IT CHECKS	PREPARE BY
Online assessment (60–90 min)	2–3 DSA problems, sometimes MCQs on CS/aptitude	Timed LC contests, HackerRank practice, read every line of constraints
DSA interview (45–60 min)	1–2 problems, communication, edge cases, complexity	Speak-aloud practice, 2 mocks/week from June (Pramp, peers)
Core CS / technical	OS, DBMS, CN, OOP, SQL, your projects in depth	The 12 questions, Feynman pages, project deep-dive script
LLD / HLD (product cos)	Structuring classes, trade-offs	6 LLD designs written in C++, 4 HLD diagrams
Behavioural / bar raiser	Ownership, conflict, failure, learning	6 STAR stories that each fit several questions
HR	Fit, relocation, notice, compensation	Research the company; know your number

Six STAR stories to prepare

1. A hard technical problem you solved (TenantWall IAM)
2. A time you failed and what you changed
3. A conflict in a team
4. Learning something fast (C++/AWS in weeks)
5. Going beyond the task (security finding in a project)
6. Leading or helping someone

Offer basics

- CTC ≠ in-hand. Ask for the split: fixed, variable, joining bonus, stocks (vesting).
- Never say a number first if you can avoid it; if pushed, give a range from levels.fyi / Glassdoor.
- Check the bond/service agreement clause before signing.
- A written offer + another interview in progress is your leverage — be polite and honest.

Mock schedule (from 31 May)

MON	TUE	WED	THU	FRI	SAT	SUN
DSA mock (peer)	Company-tagged set	Core CS mock	LLD design	OA simulation 90 min	LC + CF contest	Review + STAR practice

11

THEY DID IT FROM TIER-3

Stories ↑ Index

SDE offers and internships, from public interview experiences.

01 FULL-TIME · 2024 · SELF-REPORTED

Tier-3 → Microsoft, Google and Meta

Rejected by Google 3 times and Meta twice before getting selected at both.

Copy this: Treat rejection as data: log which round failed and fix that round only. Persistence over several cycles is normal, not a sign you're not good enough.

LeetCode Discuss

02 INTERNSHIP OFFER · AUG 2025 · SELF-REPORTED

Tier-3 → Razorpay SDE intern

Came through the college T&P cell; the drive had an 8+ CGPA cut-off.

Copy this: CGPA still opens doors at fintechs. Keep yours above 8 if you can — it's cheaper than any course.

LeetCode Discuss

03 OFF-CAMPUS · DEC 2025 · SELF-REPORTED

Tier-3 → Eternal (Zomato) SDE intern + FTE

Got in through a referral and cleared the intern + full-time process.

Copy this: Referrals are the strongest off-campus channel. Ten referral messages a week is your quota.

LeetCode Discuss

04 OFFER · MAY 2026 · SELF-REPORTED

Agoda SDE intern, 6 months, Gurugram

A six-month SDE internship offer, documented round by round.

Copy this: Six-month internships (Jan–Jun of final year) are a real route to a PPO and avoid the summer NOC problem.

LeetCode Discuss

05 SELECTED · SELF-REPORTED

Amazon SDE intern, off-campus

An off-campus Amazon internship: online assessment, then DSA interviews.

Copy this: Amazon tests DSA + Leadership Principles. Pair every DSA mock with one STAR story.

GeeksforGeeks

06 INTERNSHIP · SELECTED · SELF-REPORTED

Myntra RampUp, via Unstop

Entered through an Unstop hiring programme rather than a referral.

Copy this: Hiring contests on Unstop skip the resume filter. Enter every relevant one.

Medium

07 SELECTED · SELF-REPORTED

INDmoney SDE intern, off-campus

A fintech startup internship with DSA and project rounds.

Copy this: Startups ask about your projects in depth — have numbers ready.

GeeksforGeeks

08 SELECTED · SELF-REPORTED

Scaler SDE intern, off-campus

Ed-tech internship through an off-campus process.

Copy this: Mid-size companies are reachable earlier — use them as interview practice that might also be an offer.

GeeksforGeeks

WHAT EVERY STORY HAS IN COMMON

(1) Hundreds of problems solved before the first offer. (2) One channel that beat the college filter: a referral, a contest, a CTF, open source. (3) At least one rejection first. The college name decided none of them.

12 THEY DID IT FROM TIER-3

Stories (part 2) ↑ Index

More routes — including security, open source and the honest failures.

09 FTE + INTERN · OFF-CAMPUS · JAN 2022 · SELF-REPORTED

Accolite Digital, Mumbai

An off-campus offer from a Mumbai engineering services company.

Copy this: Mumbai-based companies are worth a separate list — no relocation question, and local alumni refer more readily.

LeetCode Discuss

10 FULL-TIME · SELF-REPORTED

Tier-3 → tier-1 software company

A personal account of moving from a tier-3 college to a top product company.

Copy this: Read the whole story for the timeline: the gap between 'started' and 'offer' is usually 12–24 months. You're on that clock now.

Medium

11 OPEN SOURCE · SELF-REPORTED

Tier-3 MCA → Google Summer of Code 2026

Got into GSoC 2026 as a tier-3 MCA student.

Copy this: GSoC is a brand that removes the college question. Start contributing in Jan; applications are usually Mar–Apr.

DEV

12 SECURITY INTERNSHIP · COMPANY PAGE

B.Tech → KPMG India cyber (CDIR) intern via CTF

A KPMG campus CTF was central to moving from final-year B.Tech student to intern on KPMG India's cyber incident response team.

Copy this: CTF results are a hiring channel for security roles. Play one a month from Feb and write up what you solve.

KPMG India

13 SECURITY CAREER · 2015 · INTERVIEW (OLDER)

Bug bounties → security engineer

Anand Prakash started reporting bugs to Facebook and others, which led to a security engineering career.

Copy this: Accepted bug reports are proof no college name can match. One valid report is worth a line at the top of your resume.

YourStory

14 CONTRACT ROLE · NEWS REPORT

Reported CBSE portal flaws → IIT Kanpur cyber team

A teenager who reported security flaws in CBSE's marks portal was taken on as an engineer in IIT Kanpur's cybersecurity team (contractual).

Copy this: Responsible disclosure, written up publicly, can be a résumé on its own.

Siasat

15 SELF-TAUGHT CAREER · COMPANY BLOG

Biology graduate → cybersecurity

Anugrah S.R. moved from biology into cybersecurity through self-learning.

Copy this: If a non-CS graduate can do it, a CSE (Cyber) student with a plan is already ahead.

CloudSEK

16 NOT SELECTED · SELF-REPORTED

Two honest rejections: Microsoft and Cisco

Two tier-3 candidates wrote up interviews they failed.

Copy this: Read failures as carefully as wins: most write-ups point to communication, edge cases or weak Core CS — all fixable before Aug 2027.

Medium + LeetCode

HOW TO READ THESE

Most are self-reported posts; outcomes aren't independently checked. The Cisco rejection is at LeetCode Discuss. Aggregator pages that list tier-3 success stories without sources (for example jobrise.io) are left out of the cards. I found no published tier-3 stories for security-engineer full-time offers specifically — the security routes above are CTFs, bug bounties and disclosure.

13 CHECK YOURSELF

Milestone checklists

↑ Index

Tick these on paper. If a box is empty at its date, the next month fixes that box first.

13 Dec 2026

- [ ] C++ STL fluent
- [ ] Striver steps 1–6 done (~150 problems)
- [ ] OS notes complete
- [ ] PortSwigger Apprentice labs done
- [ ] AWS weeks 1–4 + billing alarm
- [ ] Error log + 1-3-7-21 sheet in use

30 Jan 2027

- [ ] Striver core done (~300)
- [ ] LC contest every week
- [ ] DBMS + SQL 50 done
- [ ] TenantWall: trail, metric filter, alarm, SNS live
- [ ] AI: first RAG demo
- [ ] Resume v1

21 Mar 2027

- [ ] Hard pass 1 done
- [ ] CF Div 2: solve A–C regularly
- [ ] CN + OOP notes complete
- [ ] TenantWall paper submitted
- [ ] 1 CTF write-up
- [ ] 20 off-campus applications sent

30 May 2027

- [ ] ~450 problems, 120 Hards
- [ ] AWS SAA-C03 passed
- [ ] 4 LLD + 2 HLD designs
- [ ] 2 deployed projects with numbers
- [ ] Binary taster: 5 pwn.college levels
- [ ] Resume v4 reviewed by 2 seniors

31 Jul 2027

- [ ] ~550 problems, LC 1800+
- [ ] 16 mocks done
- [ ] 6 STAR stories rehearsed
- [ ] Company-tagged lists for top 15 targets
- [ ] 100 applications + 40 referral asks
- [ ] LinkedIn + GitHub cleaned up

Any week

- [ ] ≥ 30 problems solved
- [ ] ≥ 1 contest
- [ ] ≥ 3 h Core CS
- [ ] ≥ 5 h cyber
- [ ] ≥ 3.5 h AWS
- [ ] Sunday review done

MISTAKES THAT COST TIER-3 STUDENTS OFFERS

Watching solutions instead of solving · 1000 problems with no revision · no speaking practice · projects copied from tutorials · applying only on campus · waiting to "be ready" before applying · ignoring CGPA · quitting after the first rejection.

14 SOURCES

Everything linked in this guide

↑ Index

All free unless noted. Story links are in sections 11–12.

Courses and practice

Striver A2Z DSA sheet
https://takeuforward.org/strivers-a2z-dsa-course/strivers-a2z-dsa-course-sheet-2/

NeetCode practice
https://neetcode.io/practice

CSES problem set
https://cses.fi/problemset/

cp-algorithms
https://cp-algorithms.com

USACO Guide
https://usaco.guide

roadmap.sh
https://roadmap.sh

LeetCode SQL 50
https://leetcode.com/studyplan/top-sql-50/

System Design Primer
https://github.com/donnemartin/system-design-primer

Awesome LLD
https://github.com/ashishps1/awesome-low-level-design

PortSwigger Academy
https://portswigger.net/web-security

pwn.college
https://pwn.college

OverTheWire
https://overthewire.org/wargames/bandit/

picoCTF
https://picoctf.org

CTFtime
https://ctftime.org

flaws.cloud
https://flaws.cloud

OWASP LLM Top 10
https://genai.owasp.org/llm-top-10/

Stat 110
https://projects.iq.harvard.edu/stat110

Guides read for this edition

Tier-3 roadmap (Substack)
https://devshreebharatia.substack.com/p/tier-3-college-roadmap-for-software

Product companies from tier-3 (Medium)
https://medium.com/@hardiktaneja_99752/how-to-get-placed-in-product-based-companies-from-a-tier-3-college-f24bb4861b73

10 LPA from tier-3 (lets-code)
https://www.lets-code.co.in/blogs/how-to-get-a-10-lpa-job-as-a-fresher-from-a-tier-3-college/

HashedIn SDE-1 experience
https://leetcode.com/discuss/post/7620735/hashedin-by-deloitte-sde-1-interview-exp-urnm/

Off-campus backend internship
https://leetcode.com/discuss/post/7703521/title-sde-internship-interview-experienc-sbqj/

Intuit SDE intern (not selected)
https://www.geeksforgeeks.org/interview-experiences/intuit-interview-experience-for-sde-internship-off-campus/

Cyber jobs in India (talk)
https://sessionize.com/s/aseem-shrey/getting-a-cybersecurity-job-in-india/41330

Climb one rung at a time. The college name is a filter for the first round only — everything after it is the work in this guide.

Index

Prep Cheat Sheet. Tap a line to jump; ↑ Index on each section brings you back.

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C++ and the STL 25
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DP recipe 27
Core CS 28
SQL 28
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AWS 30
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PLACEMENT PREP · ONE PAGE

Prep cheat sheet

The rules, the triggers and the templates. Read the first section before every session, and the whole page before every test.

READ BEFORE EVERY SESSION

The rules ↑ Index

The problem loop

1. **Restate** the input, output and constraints. One example by hand. (3 min)
2. **Brute force** and its Big-O. (5 min)
3. **Optimise:** sort? hash? two pointers? binary search the answer? repeated work? a graph? (10 to 20 min)
4. **Code** it in one go. (10 to 15 min)
5. **Test** the sample and two edge cases. (5 min)
6. **Log** the pattern, the time, S/H/F and the mistake. (1 min)

When stuck

- 30 minutes alone, then a hint only
- 60 minutes, then the solution: close it and code from memory
- Re-solve at 1, 3, 7 and 21 days
- Mark it S (solved), H (hint) or F (failed)

Never zero

- Minimum day: revision, 2 Striver, 1 LeetCode, 20 min of flashcards
- Short on time, cut: HFT, CP-31, AI, then cyber. Never DSA or revision.

Edge cases to try

- Empty input, one element, two elements
- All equal, already sorted, reverse sorted
- Negative numbers and zero
- The largest n and the largest values (overflow)
- Duplicates; the target missing
- Trees: empty, one node, a straight line
- Graphs: disconnected, a self-loop, a cycle

CONSTRAINTS NAME THE ALGORITHM

Complexity ↑ Index

N UP TO	AIM FOR	TYPICAL TECHNIQUE
10 to 12	$O(n!)$	Permutations, backtracking
20 to 25	$O(2^n)$	Bitmask, subsets, meet in the middle
100	$O(n^3)$ or $O(n^4)$	Floyd-Warshall, interval DP
1,000 to 5,000	$O(n^2)$	2D DP, nested loops
10^5	$O(n \log n)$	Sorting, heap, binary search, segment tree
10^6 to 10^7	$O(n)$	Two pointers, hashing, prefix sums, sieve
10^9 to 10^{18}	$O(\log n)$ or $O(\sqrt{n})$	Binary search, maths, fast power

- About **10^8 simple operations per second**
- `int` holds about $\pm 2.1 \times 10^9$. Anything bigger: `long long` (about $\pm 9.2 \times 10^{18}$)
- Overflow happens in the expression: write `1LL * a * b`
- Mid without overflow: `lo + (hi - lo) / 2`

- Modulo: `((a % M) + M) % M` for negatives; `M = 1e9 + 7`
- Recursion deeper than about 10^5 risks a stack overflow
- `sort` is O(n log n); `map` and `set` are O(log n) per operation; `unordered_map` is O(1) on average
- A loop that halves or doubles runs log n times; `i*i <= n` runs √n times

---

WHAT YOU NEED FROM C++

C++ and the STL

↑ Index

STARTER

```
#include <bits/stdc++.h>
using namespace std;
typedef long long ll;

int main() {
    ios::sync_with_stdio(false); cin.tie(nullptr);
    int n; cin >> n;
    vector<int> a(n);
    for (int &x : a) cin >> x;
    cout << accumulate(a.begin(), a.end(), 0LL) << "\n";
}
```

CONTAINER	USE IT FOR	KEY CALLS
`vector<T>`	Dynamic array	`push_back pop_back back size [i]`
`string`	Text	`size substr(i,len) find +=`
`pair<A,B>`	Two values	`.first .second`
`stack<T>`	Last in, first out	`push pop top empty`
`queue<T>`	First in, first out (BFS)	`push pop front empty`
`deque<T>`	Both ends	`push_front push_back pop_front pop_back`
`priority_queue<T>`	Max-heap	`push pop top`
`set<T>`	Sorted unique	`insert erase count lower_bound`
`map<K,V>`	Sorted key to value	`m[k] count find erase`
`unordered_map<K,V>`	Hash map, O(1) average	`m[k] count find`

IDIOMS

```
sort(a.begin(), a.end());                       // ascending
sort(a.begin(), a.end(), greater<int>());       // descending
sort(v.begin(), v.end(), [](auto &x, auto &y){  // custom
    return x.second > y.second; });
reverse(a.begin(), a.end());
a.erase(unique(a.begin(), a.end()), a.end());   // after sort: dedupe
int i = lower_bound(a.begin(), a.end(), x) - a.begin(); // first >= x
int j = upper_bound(a.begin(), a.end(), x) - a.begin(); // first >  x
priority_queue<int, vector<int>, greater<int>> minh;    // min-heap
for (auto &[k, v] : m) { }                      // iterate a map
if (m.count(k)) { }       // check without inserting (m[k] inserts!)
__builtin_popcount(x);    // set bits (use ...ll for long long)
string t = to_string(42); int v = stoi("42");
```

- `top()` or `pop()` on an empty container is undefined: check `empty()` first
- `v.size()` is unsigned: `v.size() - 1` underflows when empty
- Pass big things by reference: `const vector<int> &v`
- After `cin >> n`, call `cin.ignore()` before `getline`

THE TRIGGER TELLS YOU THE TECHNIQUE

Pattern triggers

↑ Index

IF THE PROBLEM SAYS…	THINK	WHY
Sorted array, find a target or a boundary	Binary search	Halve the search space each step
"Minimum of the maximum", "smallest value such that…"	Binary search on the answer	The answer is monotonic: check(x) flips once
Pair or triplet with a sum, in a sorted array	Two pointers	Move the end that helps
Longest or shortest subarray or substring with a condition	Sliding window	Grow right, shrink left while invalid
Subarray sum equals K, count of subarrays	Prefix sum + hash map	sum(l..r) = pre[r] – pre[l–1]
Next greater or smaller element, histogram areas	Monotonic stack	Each element is pushed and popped once
Top K, Kth largest, merge K sorted	Heap	Keep only K items
Seen before? count? group by?	Hash map or set	O(1) lookups
All subsets, permutations, combinations	Backtracking	Choose, explore, un-choose
Shortest path, unweighted	BFS	Level by level
Shortest path, weights ≥ 0	Dijkstra	Greedy on the closest node
Negative edges; all pairs	Bellman-Ford; Floyd-Warshall	Relax edges repeatedly
Connected groups, "same component", merging	DFS or Union-Find (DSU)	Near O(1) per union
Order with prerequisites	Topological sort	Kahn's algorithm with in-degrees
Count ways, min or max cost, "can you reach"	Dynamic programming	Overlapping subproblems
Choices now affect later, local best works	Greedy (sort first)	Prove it with an exchange argument
Prefix lookups, autocomplete, max XOR	Trie	One node per character or bit
Appears once, toggle, subsets up to 20	Bit manipulation	XOR cancels pairs; masks enumerate subsets
Linked list: middle, cycle	Slow and fast pointers	Fast moves two, slow moves one
Tree: any "path" or "height" question	DFS returning a value up	Combine the left and right answers
Intervals: merge, overlap, rooms	Sort by start or end	Then sweep once

TYPE THESE UNTIL THEY ARE AUTOMATIC

Templates

↑ Index

BINARY SEARCH

```
int lo = 0, hi = n - 1, ans = -1;
while (lo <= hi) {
    int mid = lo + (hi - lo) / 2;
    if (a[mid] >= x) { ans = mid; hi = mid - 1; }   // first index >= x
    else lo = mid + 1;
}
```

BINARY SEARCH ON THE ANSWER

```
ll lo = MIN_ANS, hi = MAX_ANS, ans = hi;
while (lo <= hi) {
    ll mid = lo + (hi - lo) / 2;
    if (ok(mid)) { ans = mid; hi = mid - 1; }   // smallest value that works
    else lo = mid + 1;
}
```

TWO POINTERS (SORTED ARRAY)

```
int i = 0, j = n - 1;
while (i < j) {
    int s = a[i] + a[j];
    if (s == target) { /* found */ i++; j--; }
    else if (s < target) i++;
    else j--;
}
```

SLIDING WINDOW (LONGEST VALID)

```
int best = 0;
for (int l = 0, r = 0; r < n; r++) {
    add(a[r]);
    while (!valid()) remove(a[l++]);
    best = max(best, r - l + 1);
}
```

PREFIX SUM; SUBARRAYS WITH SUM K

```
vector<ll> pre(n + 1, 0);
for (int i = 0; i < n; i++) pre[i + 1] = pre[i] + a[i];
// sum of a[l..r] = pre[r + 1] - pre[l]

unordered_map<ll, int> seen; seen[0] = 1;
ll sum = 0, ways = 0;
for (int x : a) { sum += x; ways += seen[sum - k]; seen[sum]++; }
```

MONOTONIC STACK (NEXT GREATER)

```
vector<int> nge(n, -1); stack<int> st;       // st holds indices
for (int i = 0; i < n; i++) {
    while (!st.empty() && a[st.top()] < a[i]) { nge[st.top()] = a[i]; st.pop();
}
    st.push(i);
}
```

BACKTRACKING (SUBSETS)

```
void go(int i, vector<int> &cur) {
    if (i == n) { out.push_back(cur); return; }
    cur.push_back(a[i]); go(i + 1, cur);     // choose
    cur.pop_back();      go(i + 1, cur);     // un-choose
}
```

BFS (SHORTEST PATH, UNWEIGHTED)

```
vector<int> dist(n, -1); queue<int> q;
dist[s] = 0; q.push(s);
while (!q.empty()) {
    int u = q.front(); q.pop();
    for (int v : adj[u]) if (dist[v] == -1) { dist[v] = dist[u] + 1; q.push(v);
}
}
```

DFS; GRID NEIGHBOURS

```
void dfs(int u) {
    vis[u] = true;
    for (int v : adj[u]) if (!vis[v]) dfs(v);
}
int dr[] = {-1, 1, 0, 0}, dc[] = {0, 0, -1, 1};
// nr = r + dr[k], nc = c + dc[k]; check 0 <= nr < R and 0 <= nc < C
```

TOPOLOGICAL SORT (KAHN)

```
vector<int> indeg(n), order; queue<int> q;
for (int u = 0; u < n; u++) for (int v : adj[u]) indeg[v]++;
for (int u = 0; u < n; u++) if (!indeg[u]) q.push(u);
while (!q.empty()) {
    int u = q.front(); q.pop(); order.push_back(u);
    for (int v : adj[u]) if (--indeg[v] == 0) q.push(v);
}
// order.size() < n  means there is a cycle
```

DIJKSTRA

```
vector<ll> d(n, LLONG_MAX);
priority_queue<pair<ll,int>, vector<pair<ll,int>>, greater<>> pq;
d[s] = 0; pq.push({0, s});
while (!pq.empty()) {
    auto [du, u] = pq.top(); pq.pop();
    if (du > d[u]) continue;
    for (auto [v, w] : adj[u])
        if (du + w < d[v]) { d[v] = du + w; pq.push({d[v], v}); }
}
```

UNION-FIND (DSU)

```
vector<int> p(n), sz(n, 1);
iota(p.begin(), p.end(), 0);
int find(int x) { return p[x] == x ? x : p[x] = find(p[x]); }
bool unite(int a, int b) {
    a = find(a); b = find(b);
    if (a == b) return false;
    if (sz[a] < sz[b]) swap(a, b);
    p[b] = a; sz[a] += sz[b]; return true;
}
```

DP: 0/1 KNAPSACK, LIS, LCS

```
// knapsack: best value with capacity W
vector<int> dp(W + 1, 0);
for (int i = 0; i < n; i++)
    for (int c = W; c >= wt[i]; c--) dp[c] = max(dp[c], dp[c - wt[i]] +
val[i]);

// LIS in O(n log n)
vector<int> t;
for (int x : a) { auto it = lower_bound(t.begin(), t.end(), x);
                  if (it == t.end()) t.push_back(x); else *it = x; }

// LCS of s and r
vector<vector<int>> L(n + 1, vector<int>(m + 1, 0));
for (int i = 1; i <= n; i++) for (int j = 1; j <= m; j++)
    L[i][j] = s[i-1] == r[j-1] ? L[i-1][j-1] + 1 : max(L[i-1][j], L[i][j-1]);
```

LINKED LIST: REVERSE; SLOW AND FAST

```
Node *prev = nullptr, *cur = head;
while (cur) { Node *nx = cur->next; cur->next = prev; prev = cur; cur = nx; }
head = prev;

Node *slow = head, *fast = head;             // middle, or cycle check
while (fast && fast->next) { slow = slow->next; fast = fast->next->next;
                             if (slow == fast) { /* cycle */ break; } }
```

TREE: HEIGHT, LEVEL ORDER

```
int height(Node *r) { return r ? 1 + max(height(r->left), height(r->right)) :
0; }

queue<Node*> q; if (root) q.push(root);
while (!q.empty()) {
    int k = q.size();                        // one level
    while (k--) { Node *u = q.front(); q.pop();
                  if (u->left) q.push(u->left); if (u->right) q.push(u->right);
}
}
```

DYNAMIC PROGRAMMING IN FIVE QUESTIONS

DP recipe

↑ Index

1. **State:** what changes between subproblems? (index, capacity, last choice)
2. **Meaning:** say dp[i] in one sentence
3. **Transition:** the choices at this state
4. **Base case:** the smallest inputs
5. **Order and answer:** which way to fill, and where the answer sits

FAMILY	STATE	EXAMPLES
1D	dp[i]	Climbing stairs, house robber

FAMILY	STATE	EXAMPLES
Grid	dp[r][c]	Unique paths, minimum path sum
Subsequence	dp[i][sum]	Subset sum, knapsack, coin change
Two strings	dp[i][j]	LCS, edit distance
LIS	dp[i] ending at i	LIS, longest chain
Stocks	dp[day][holding][k]	Buy and sell variants
Intervals	dp[l][r]	Matrix chain, burst balloons

ONE-LINE ANSWERS, THEN GO DEEPER

Core CS ↑ Index

OOP

- **Encapsulation:** data and methods together, internals hidden
- **Abstraction:** show what it does, hide how
- **Inheritance:** a class reuses and extends another
- **Polymorphism:** one interface, many forms. Compile-time: overloading. Run-time: virtual functions through the vtable.
- A **virtual destructor** is needed when deleting through a base pointer
- **Shallow copy** copies pointers; **deep copy** copies what they point to
- **SOLID:** single responsibility, open/closed, Liskov, interface segregation, dependency inversion

Operating systems

- **Process** has its own memory; **threads** share it
- **Context switch:** save one state, load another
- **Scheduling:** FCFS, SJF, round robin, priority
- **Deadlock** needs all four: mutual exclusion, hold and wait, no preemption, circular wait
- **Mutex** has one owner; a **semaphore** is a counter
- **Paging** maps virtual pages to frames; a page fault loads from disk; thrashing is constant faulting
- **Virtual memory** lets processes use more than physical RAM

DBMS

- **ACID:** atomicity, consistency, isolation, durability
- **Normal forms:** 1NF atomic values; 2NF no partial dependency; 3NF no transitive dependency; BCNF every determinant is a key
- **Keys:** primary, candidate, foreign, composite
- **Index:** a B+ tree that speeds reads and slows writes
- **Isolation levels:** read uncommitted, read committed, repeatable read, serializable
- **Joins:** inner, left, right, full, cross, self

Networks

- **OSI:** physical, data link, network, transport, session, presentation, application
- **TCP:** reliable, ordered, 3-way handshake (SYN, SYN-ACK, ACK). **UDP:** no guarantees, faster
- **Typing a URL:** DNS lookup, TCP handshake, TLS handshake, HTTP request, response, render
- **HTTP codes:** 2xx ok, 3xx redirect, 4xx your fault, 5xx server fault
- **Subnetting:** /24 has 256 addresses, 254 hosts
- **NAT** maps private to public addresses; **DHCP** hands out addresses; **ARP** maps IP to MAC

WRITE THE CLAUSES IN THIS ORDER

SQL ↑ Index

SHAPE OF A QUERY

```
SELECT dept, COUNT(*) AS n, AVG(salary) AS avg_sal
FROM employees e
JOIN departments d ON d.id = e.dept_id
WHERE e.active = 1            -- filters rows
GROUP BY dept
HAVING COUNT(*) > 5           -- filters groups
```

```
ORDER BY avg_sal DESC
LIMIT 10;
-- runs as: FROM, JOIN, WHERE, GROUP BY, HAVING, SELECT, ORDER BY, LIMIT
```

THE INTERVIEW FAVOURITES

```
-- second highest salary
SELECT MAX(salary) FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);

-- top earner in each department
SELECT * FROM (
  SELECT e.*, DENSE_RANK() OVER (PARTITION BY dept_id ORDER BY salary DESC) AS r
  FROM employees e) t
WHERE r = 1;

-- duplicates
SELECT email, COUNT(*) FROM users GROUP BY email HAVING COUNT(*) > 1;

-- running total
SELECT day, SUM(amount) OVER (ORDER BY day) AS running FROM sales;

-- a CTE
WITH big AS (SELECT * FROM orders WHERE total > 1000)
SELECT customer_id, COUNT(*) FROM big GROUP BY customer_id;
```

LABS ONLY: PLATFORMS BUILT FOR IT AND MACHINES YOU OWN

Security ↑ Index

LINUX

```
ls -la; pwd; cd; cat; less; head; tail -f
grep -rn "text" .        find / -name "*.conf" 2>/dev/null
chmod 640 file           chown user:group file
ps aux; top; kill -9 PID
ss -tulpn                # listening ports
sudo -l                  # what may I run as root?
find / -perm -4000 2>/dev/null   # SUID files
curl -I https://site     # response headers
```

RECON AND CAPTURE

```
nmap -sV -sC -p- target       # versions, default scripts, all ports
nmap -sn 192.168.1.0/24       # which hosts are up
tcpdump -i eth0 port 80
# Wireshark filters
http; dns; tcp.port == 443; ip.addr == 10.0.0.5
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

PORT	SERVICE
20, 21	FTP
22	SSH
23	Telnet
25	SMTP
53	DNS
80 / 443	HTTP / HTTPS
110 / 143	POP3 / IMAP
3306	MySQL
3389	RDP
5432	PostgreSQL

OWASP Top 10 (2021)

1. Broken access control (IDOR)
2. Cryptographic failures
3. Injection (SQL, command, XSS)
4. Insecure design
5. Security misconfiguration
6. Vulnerable and outdated components
7. Identification and authentication failures
8. Software and data integrity failures
9. Logging and monitoring failures
10. Server-side request forgery (SSRF)

- **Fix injection** with parameterised queries, never string building
- **Fix XSS** with output encoding and a content security policy
- **Hash passwords** with bcrypt or Argon2 and a salt; hashing is one-way, encryption is not
- **STRIDE:** spoofing, tampering, repudiation, information disclosure, denial of service, elevation of privilege

---

FOR THE SAA EXAM AND FOR TENANTWALL

AWS ↑ Index

SERVICE	WHAT IT IS
IAM	Users, groups, roles, policies. Who can do what.
EC2 / EBS	Virtual servers / their disks
S3	Object storage in buckets
VPC	Your private network: subnets, route tables, gateways
RDS / DynamoDB	Managed SQL / NoSQL key-value
ELB + Auto Scaling	Spread load / add and remove servers
Route 53	DNS
CloudFront	CDN at edge locations
Lambda	Run code without servers
SQS / SNS	Queue / publish and subscribe
CloudTrail	Who called which API, when
CloudWatch	Metrics, logs, alarms
KMS	Encryption keys
CloudFormation	Infrastructure as code

IAM, the part TenantWall lives on

- **Evaluation order:** an explicit Deny wins; otherwise an Allow is needed; otherwise implicit deny
- **Identity policy** attaches to a user, group or role; **resource policy** attaches to the thing (a bucket)
- **ABAC:** compare `aws:PrincipalTag/Team` with `aws:ResourceTag/Team`
- **Least privilege:** grant only what is needed
- **Role:** temporary credentials, no long-lived keys

Remember for the exam

- **Security group:** stateful, allow rules only, on the instance. **NACL:** stateless, allow and deny, on the subnet.
- **Public subnet** routes to an internet gateway; **private** reaches out through a NAT gateway
- **Multi-AZ** is for availability; **read replicas** are for read scaling
- **S3 classes:** Standard, Infrequent Access, Glacier, by how often you read
- **Shared responsibility:** AWS secures the cloud; you secure what you put in it

---

THE NIGHT BEFORE AND IN THE ROOM

Interviews ↑ Index

In a coding round

1. Repeat the problem; ask for constraints and an example
2. Say the brute force and its complexity
3. Improve it out loud; get a nod before coding
4. Code cleanly, naming things by meaning
5. Walk one example through the code
6. State time and space; mention edge cases

Project walk-through, 3 minutes

1. The problem, in one sentence
2. What you built and your part in it
3. One hard decision and why you made it
4. How you tested it, with a number
5. What you would change next

TenantWall: 2 identity policies for any number of teams; 14 of 14 tests; refusals logged and alerted.

Five stories to have ready

- A hard bug you fixed
- A disagreement in a team
- A deadline you nearly missed
- A failure and what changed after
- Something you taught yourself

Tell each as situation, task, action, result. End every interview with one real question.

Templates are C++17. Type them yourself until they are automatic; do not paste them. 13 Nov 2026

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Placement Roadmap. Sections, then every week and month of the plan. Tap to jump.

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RAIT · CSE (CYBERSECURITY) · CLASS OF 2028

An SDE with cloud-security depth ↑ Index

One plan from November 2026 to graduation. DSA comes first, because coding tests are the first filter everywhere. Cloud, security and AI are built on top, so that by final placements you have four connected projects, a paper and a certification.

● NOW · UNTIL 15 NOV

Village week

On your phone, about an hour a day: Learn C++ at 2× (slow down only for what differs from C), then Time complexity and the two Patterns lessons. Write code in a notebook.

Next: C++ and STL, from 16 Nov 2026 (3 days away)

PLACEMENT SEASON OPENS

261 days, on 1 Aug 2027

NEXT MILESTONE

Gate test passed

in 4 days, 17 Nov 2026

TARGETS

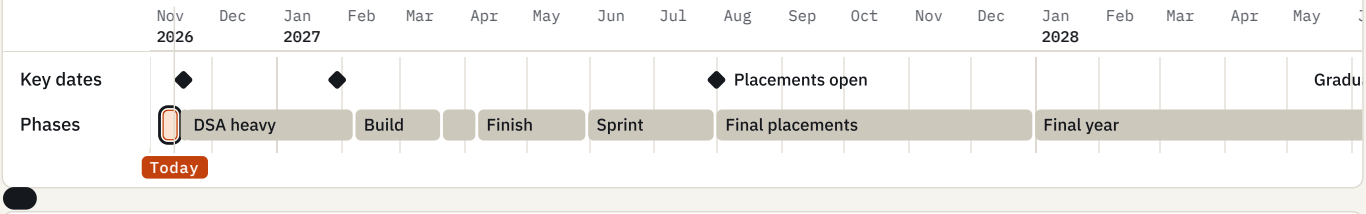
Main: SDE at a product company
Second: cloud or security engineer
Stretch: SDE at an HFT firm

NOVEMBER 2026 TO JUNE 2028

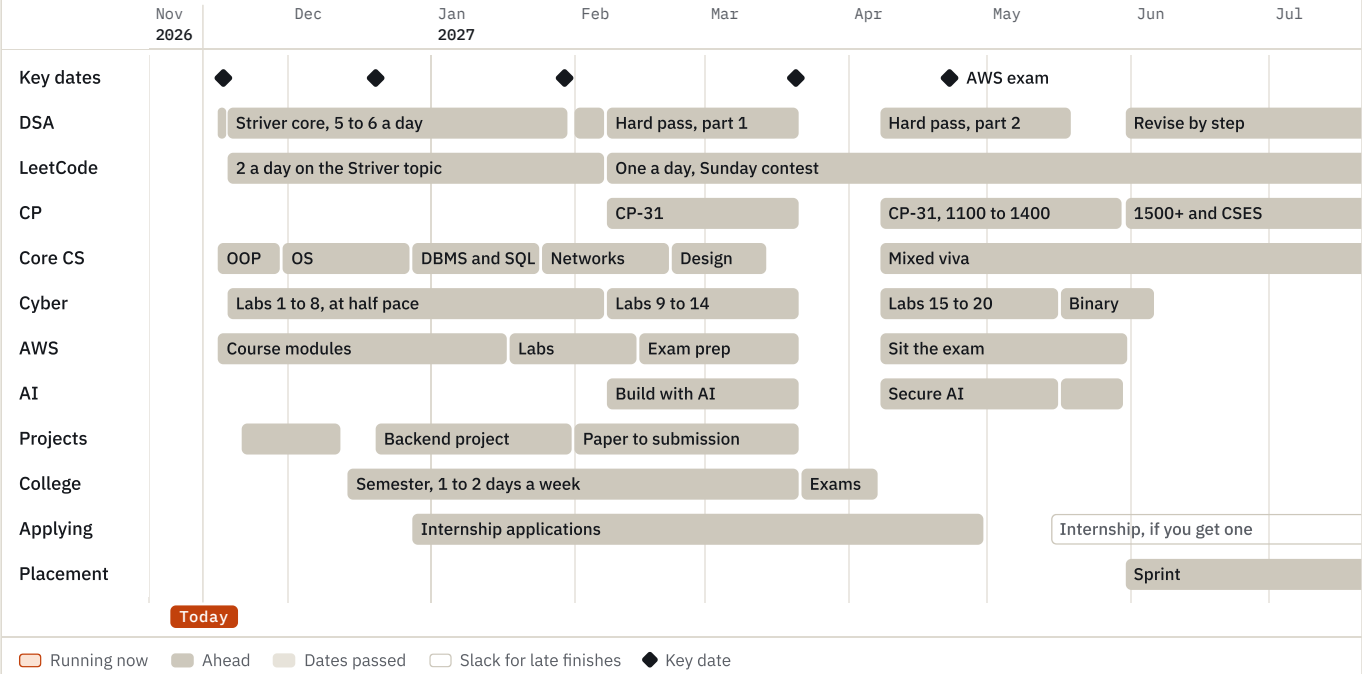
Timeline ↑ Index

The first chart is the whole journey. The second zooms into the plan itself. Select a bar to see what it covers. On a phone, swipe a chart sideways.

The whole journey Nov 2026 to Jun 2028



The first nine months, track by track Nov 2026 to Jul 2027



Village week

7 Nov to 15 Nov 2026 · About 1 h a day, on your phone

On your phone, about an hour a day: Learn C++ at 2× (slow down only for what differs from C), then Time complexity and the two Patterns lessons. Write code in a notebook.

NINE PHASES

What each phase is for ↑ Index

Dates follow the intensive track. Striver core can slip to the end of February and the Hard pass, AI and AWS exam to the end of June and still be on time, but there is no slack beyond that.

7 Nov to 15 Nov 2026
About 1 h a day, on your phone

Village week • NOW

- Learn C++ at 2×: cin and cout, string, references, vector, auto
- Time complexity, both Patterns lessons, Flowcharts and pseudocode, Dry runs and debugging
- On paper: the pattern tables and the five recursions
- Read OOP in the playbook, Part 04

Done when the videos are watched and the notebook has the patterns and recursions.

16 Nov to 17 Nov 2026
Two full days

C++ and STL NEXT

- Day 1: drills, D1-1 to D1-10, start the STL video
- Day 2: STL drills, recursion and complexity, gate test at 7 PM with no notes or AI
- AWS starts at 30 minutes a day: billing alarm, then IAM
- On 18 Nov, export your progress and move to the offline copy. Claude Pro ends on the 19th.

Done when the gate test is 4 of 5.

18 Nov 2026 to 7 Feb 2027
About 9 h a day, 6 days a week

DSA heavy

- Striver 4 h: all 359 Easy and Medium problems by 30 Jan, then a week of redoing every H and F
- LeetCode 1.5 h · Core CS 1 h (OOP, OS, DBMS, networks) · AWS 30 min · cyber 30 min
- TenantWall on four Saturdays to 12 Dec · the backend project on Sundays from 20 Dec to 31 Jan
- From 14 Dec: college 1 to 2 days a week · from 28 Dec: 5 internship applications a week · from 4 Jan: 15 minutes of aptitude · from mid-January: mock interviews

Done when Striver core is done, the backend project is deployed, and resume version 2 is out.

8 Feb to 21 Mar 2027
About 8 h a day

Build

- Hard pass, part 1: 42 Hard problems, one a day
- AI weeks 1 to 6: LLM basics through to a shipped, tested AI feature
- CP-31 from rating 800 to 1000 · cyber labs 9 to 14 · system design
- AWS exam prep · internship applications · TenantWall paper submitted by 21 Mar

Done when the paper is submitted and the first AWS practice exam is done.

22 Mar to 7 Apr 2027
Minimum day only

Semester exams

- Revision 30 min, 2 problems, 1 LeetCode problem
- Everything else is paused and restarts on 8 Apr

Done when exams are over. Rest for two days.

8 Apr to 30 May 2027
About 8 h a day

Finish and certify

- Hard pass, part 2: the other 42 Hard problems, done by 19 May
- AI weeks 7 to 12: securing LLM apps, then the intrusion-detection capstone
- AWS: second practice exam, then SAA-C03 in the week of 19 Apr

- CP-31 from 1100 to 1400 · cyber labs 15 to 20, then the binary taster
- With a summer internship offer: it starts around mid-May and replaces the capstone

Done when you are certified, the Hard pass is done and the capstone has a README, a diagram and real numbers.

31 May to 31 Jul 2027
Full days

Placement sprint

- Revise Striver by step, then daily mixed timed sets
- One timed online-test simulation and one mock interview every week
- Core CS answers out loud · final resume · 40-company list · referrals
- Optional: harder CP-31 and CSES, and probability for HFT

Done when you pass most mock interviews.

1 Aug to 31 Dec 2027
Placement season

Final placements

- On-campus drives, plus 10 off-campus applications and 5 referral requests a week
- SDE first; cloud and security roles are the strong second lane
- After every test: write down the questions and fix the gap that week

Done when you hold an offer you want.

1 Jan to 30 Jun 2028
Around college

Final year

- Final-year project: extend TenantWall or the intrusion-detection work
- A second certification: AWS Security Specialty or Security+
- Keep applying off-campus if there is no offer yet

Done when you graduate in June and start work in July.

37 WEEKS, THEN 11 MONTHS · INTENSIVE TRACK

Week by week ↑ Index

Every week from 16 November 2026 to the start of placements, with what each track does. Weeks up to the end of May also open day by day, with each day's Striver or Hard-pass problems. After July 2027 the plan runs by month.

Open all weeks

C++ and STL, then DSA heavy

– WEEK 1 **16 Nov to 22 Nov 2026** GATE TEST, 7 TO 9 PM EXPORT PROGRESS; CLAUDE PRO ENDS ON THE 19TH

DSA	Mon and Tue: C++ and STL, then the gate test. Wed to Sat: Step 1 Learn the basics (24 problems).
LeetCode	From Wed: 2 problems a day on arrays and maths basics. Sunday contest.
Core CS	Object-oriented programming (C++): Classes and encapsulation. Classes and objects, members, this; Access specifiers: public, private, protected; Constructors: default, parameterised, copy; initialiser lists; Destructors and object lifetime; shallow vs deep copy; Static members and functions; const member functions.
Cyber	Lab week 1: Linux command line. TryHackMe Linux Fundamentals parts 1 to 3; OverTheWire Bandit levels 0 to 5. Proof: A notes file of 30 commands, each with one example. 30 minutes a day and 2 hours on Sunday; a lab week can take two calendar weeks.
AWS	Modules 1, 2 and 8 (IAM): users, groups, roles, policies, explicit vs implicit deny. Set a billing alarm and budget first. Feeds TenantWall's Policy Generator work.
TenantWall	Saturday, 3 hours: Policies: Policy Generator and explicit deny. Rebuild the bucket policy (Listing 3) in the AWS Policy Generator and compare it with yours; Run IAM Access Analyzer's policy check on all three policies; Write the explicit versus implicit deny subsection; mark each test in Table II.

DAY BY DAY

– Mon 16 Nov C++ day 1

C++ day 1

- Learn C++ at 2×: slow down only for what differs from C
- Day 1 drills: input and output, types and overflow, loops and patterns, functions and references, vectors and strings
- Evening: practice problems D1-1 to D1-10

– Tue 17 Nov C++ day 2

C++ day 2

- STL video, then the STL I, II and III drills
- Recursion, pointers and structs; complexity and hashing
- 7 to 9 PM: gate test, no notes or AI, pass with 4 of 5

– Wed 18 Nov Striver: step 1

Striver: step 1

- Things to know in C++: input/output, data types, if/else, switch, loops (Easy)
- Functions: pass by value vs reference, pointers basics (Easy)
- Arrays and strings in C++ (Easy)
- Patterns 1 to 11 (squares, triangles, number triangles) (Easy)
- Patterns 12 to 17 (letter and mirrored patterns) (Easy)
- Patterns 18 to 22 (hollow, diamond, butterfly) (Easy)

Core CS: Object-oriented programming (C++) · Classes and encapsulation

- Constructors: default, parameterised, copy; initialiser lists
- Answer 3 questions on it out loud

– Thu 19 Nov Striver: step 1

Striver: step 1

- STL containers: pair, vector, list, deque, stack, queue, priority_queue (Easy)
- STL containers: set, multiset, unordered_set, map, multimap, unordered_map (Easy)
- STL algorithms: sort, next_permutation, __builtin_popcount, max_element (Easy)
- Count digits (Easy)
- Reverse a number (Easy)
- Check palindrome (Easy)

Core CS: Object-oriented programming (C++) · Classes and encapsulation

- Destructors and object lifetime; shallow vs deep copy
- Answer 3 questions on it out loud

– Fri 20 Nov Striver: step 1

Striver: step 1

- GCD or HCF (Easy)
- Armstrong numbers (Easy)
- Print all divisors (Easy)
- Check for prime (Easy)
- Print name N times (Easy)
- Print 1 to N and N to 1 (Easy)

Core CS: Object-oriented programming (C++) · Classes and encapsulation

- Static members and functions; const member functions
- Answer 3 questions on it out loud

– Sat 21 Nov Striver: step 1

Striver: step 1

- Sum of first N numbers (Easy)
- Factorial of N (Easy)
- Reverse an array using recursion (Easy)
- Check palindrome using recursion (Easy)
- Fibonacci number (multiple recursion calls) (Easy)
- Hashing theory: division method, collisions, unordered_map worst case (Easy)

TenantWall, 3 h (in place of LeetCode)

- Rebuild the bucket policy (Listing 3) in the AWS Policy Generator and compare it with yours
- Run IAM Access Analyzer's policy check on all three policies
- Write the explicit versus implicit deny subsection; mark each test in Table II

Core CS: Object-oriented programming (C++) · Classes and encapsulation

- Classes and objects, members, this
- Answer 3 questions on it out loud

– Sun 22 Nov Sunday

Sunday

- Weekly contest, then upsolve one problem
- Written weekly review: solved, failed and why, what changes
- 2 hours of cyber
- Rest

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 2 23 Nov to 29 Nov 2026

DSA	Step 1 Learn the basics, Step 2 Sorting techniques, Step 3 Arrays: 36 new problems, 6 a day, Monday to Saturday.
LeetCode	2 problems a day on this week's Striver topic, 35 minutes each. Not on Saturday. Sunday contest, then upsolve one.
Core CS	Object-oriented programming (C++): Inheritance and polymorphism. Inheritance types and access in inheritance; Function overloading and operator overloading (compile-time); Virtual functions, overriding, vtable and vptr (run-time); Abstract classes, pure virtual functions, interfaces; Virtual destructors, diamond problem, virtual inheritance; Composition vs inheritance; the SOLID principles.
Cyber	Lab week 1: Linux command line. TryHackMe Linux Fundamentals parts 1 to 3; OverTheWire Bandit levels 0 to 5. Proof: A notes file of 30 commands, each with one example. 30 minutes a day and 2 hours on Sunday; a lab week can take two calendar weeks.
AWS	Module 9: elasticity, high availability and monitoring. CloudWatch logs, metrics and alarms; CloudTrail. Feeds TenantWall's trail and alarm.
TenantWall	Saturday, 3 hours: CloudTrail trail. Create tenantwall-trail with logs in a separate bucket; Turn on management events, and S3 data events for tenantwallbucket only; Re-run tenantwall_tests.py and find each refusal in the logs.

DAY BY DAY

– Mon 23 Nov Striver: step 1, step 2

Striver: step 1, step 2

- Counting frequencies of array elements (Easy)
- Highest and lowest frequency element (Easy)
- Selection sort (Easy)
- Bubble sort (Easy)
- Insertion sort (Easy)
- Merge sort (Medium)

Core CS: Object-oriented programming (C++) · Inheritance and polymorphism

- Inheritance types and access in inheritance
- Answer 3 questions on it out loud

– Tue 24 Nov Striver: step 2, step 3

Striver: step 2, step 3

- Quick sort (Medium)
- Recursive bubble sort (Easy)
- Recursive insertion sort (Easy)
- Largest element in an array (Easy)
- Second largest element without sorting (Easy)
- Check if the array is sorted (Easy)

Core CS: Object-oriented programming (C++) · Inheritance and polymorphism

- Function overloading and operator overloading (compile-time)
- Answer 3 questions on it out loud

– Wed 25 Nov Striver: step 3

Striver: step 3

- Remove duplicates from a sorted array (Easy)
- Left rotate an array by one place (Easy)
- Left rotate an array by D places (Easy)
- Move zeros to the end (Easy)
- Linear search (Easy)
- Union of two sorted arrays (Easy)

Core CS: Object-oriented programming (C++) · Inheritance and polymorphism

- Virtual functions, overriding, vtable and vptr (run-time)
- Answer 3 questions on it out loud

– **Thu 26 Nov** Striver: step 3

Striver: step 3

- Find the missing number (Easy)
- Maximum consecutive ones (Easy)
- Find the number that appears once (Easy)
- Longest subarray with sum K (positives) (Easy)
- Longest subarray with sum K (positives and negatives) (Easy)
- Two Sum (check and indices) (Medium)

Core CS: Object-oriented programming (C++) · Inheritance and polymorphism

- Abstract classes, pure virtual functions, interfaces
- Answer 3 questions on it out loud

– **Fri 27 Nov** Striver: step 3

Striver: step 3

- Sort an array of 0s, 1s and 2s (Medium)
- Majority element (more than n/2) (Medium)
- Maximum subarray sum (Kadane) (Medium)
- Print the subarray with maximum sum (Medium)
- Best time to buy and sell stock (Medium)
- Rearrange array elements by sign (Medium)

Core CS: Object-oriented programming (C++) · Inheritance and polymorphism

- Virtual destructors, diamond problem, virtual inheritance
- Answer 3 questions on it out loud

– **Sat 28 Nov** Striver: step 3

Striver: step 3

- Next permutation (Medium)
- Leaders in an array (Medium)
- Longest consecutive sequence (Medium)
- Set matrix zeroes (Medium)
- Rotate matrix by 90 degrees (Medium)
- Spiral traversal of a matrix (Medium)

TenantWall, 3 h (in place of LeetCode)

- Create tenantwall-trail with logs in a separate bucket
- Turn on management events, and S3 data events for tenantwallbucket only
- Re-run tenantwall_tests.py and find each refusal in the logs

Core CS: Object-oriented programming (C++) · Inheritance and polymorphism

- Composition vs inheritance; the SOLID principles
- Answer 3 questions on it out loud

– **Sun 29 Nov** Sunday

Sunday

- Weekly contest, then upsolve one problem
- Written weekly review: solved, failed and why, what changes
- 2 hours of cyber
- Rest

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 3 **30 Nov to 6 Dec 2026**

DSA	Step 3 Arrays, Step 4 Binary search, Step 5 Strings (basic and medium): 36 new problems, 6 a day, Monday to Saturday.
LeetCode	2 problems a day on this week's Striver topic, 35 minutes each. Not on Saturday. Sunday contest, then upsolve one.
Core CS	Operating systems: Processes and threads. What an OS does; kernel mode vs user mode; system calls; Process vs thread; the PCB; process states; Context switching and its cost; fork, exec, wait; zombie and orphan processes; Inter-process communication: pipes, shared memory, messages.
Cyber	Lab week 2: Permissions and the shell. Bandit 6 to 15; file permissions, SUID, pipes and redirection. Proof: Bandit notes 0 to 15 (what each level taught). 30 minutes a day and 2 hours on Sunday; a lab week can take two calendar weeks.
AWS	Module 3 (S3): bucket policies, encryption, versioning, lifecycle, Block Public Access.
TenantWall	Saturday, 3 hours: CloudWatch alarm. Send the trail to CloudWatch Logs; Metric filter for refused calls; alarm at 3 or more in 5 minutes, emailed through SNS; Second filter for tag tampering: CreateTags, TagUser, UntagUser.

DAY BY DAY

– **Mon 30 Nov** Striver: step 3, step 4

Striver: step 3, step 4
- Count subarrays with sum K (Medium)
- Binary search to find X (Easy)
- Implement lower bound (Easy)
- Implement upper bound (Easy)
- Search insert position (Easy)
- Floor and ceil in a sorted array (Easy)

Core CS: Operating systems · Processes and threads
- What an OS does; kernel mode vs user mode; system calls
- Answer 3 questions on it out loud

– **Tue 1 Dec** Striver: step 4

Striver: step 4
- First and last occurrence (Easy)
- Count occurrences in a sorted array (Easy)
- Search in a rotated sorted array I (Medium)
- Search in a rotated sorted array II (duplicates) (Medium)
- Minimum in a rotated sorted array (Medium)
- How many times the array is rotated (Medium)

Core CS: Operating systems · Processes and threads
- Process vs thread; the PCB; process states
- Answer 3 questions on it out loud

– **Wed 2 Dec** Striver: step 4

Striver: step 4
- Single element in a sorted array (Medium)
- Find a peak element (Medium)
- Square root of N (Easy)
- Nth root of M (Medium)
- Koko eating bananas (Medium)
- Minimum days to make M bouquets (Medium)

Core CS: Operating systems · Processes and threads
- Context switching and its cost
- Answer 3 questions on it out loud

– **Thu 3 Dec** Striver: step 4

Striver: step 4
- Smallest divisor given a threshold (Medium)
- Capacity to ship packages within D days (Medium)
- Kth missing positive number (Medium)
- Row with the maximum number of 1s (Medium)
- Search in a 2D matrix I (Medium)
- Search in a 2D matrix II (Medium)

Core CS: Operating systems · Processes and threads

- fork, exec, wait; zombie and orphan processes
- Answer 3 questions on it out loud

– Fri 4 Dec Striver: step 5

Striver: step 5

- Remove outermost parentheses (Easy)
- Reverse words in a string (Easy)
- Largest odd number in a string (Easy)
- Longest common prefix (Easy)
- Isomorphic strings (Easy)
- Check if one string is a rotation of another (Easy)

Core CS: Operating systems · Processes and threads

- Inter-process communication: pipes, shared memory, messages
- Answer 3 questions on it out loud

– Sat 5 Dec Striver: step 5

Striver: step 5

- Valid anagram (Easy)
- Sort characters by frequency (Medium)
- Maximum nesting depth of parentheses (Easy)
- Roman to integer (Easy)
- Integer to Roman (Medium)
- Implement atoi (Medium)

TenantWall, 3 h (in place of LeetCode)

- Send the trail to CloudWatch Logs
- Metric filter for refused calls; alarm at 3 or more in 5 minutes, emailed through SNS
- Second filter for tag tampering: CreateTags, TagUser, UntagUser

Core CS: Operating systems · Processes and threads

- What an OS does; kernel mode vs user mode; system calls
- Answer 3 questions on it out loud

– Sun 6 Dec Sunday

Sunday

- Weekly contest, then upsolve one problem
- Written weekly review: solved, failed and why, what changes
- 2 hours of cyber
- Rest

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 4 7 Dec to 13 Dec 2026

DSA	Step 5 Strings (basic and medium), Step 6 Linked list, Step 7 Recursion and backtracking: 36 new problems, 6 a day, Monday to Saturday.
LeetCode	2 problems a day on this week's Striver topic, 35 minutes each. Not on Saturday. Sunday contest, then upsolve one.
Core CS	Operating systems: CPU scheduling. FCFS, SJF, SRTF, priority, round robin; Waiting, turnaround and response time; Gantt charts; Starvation and aging; Choosing a time quantum for round robin.
Cyber	Lab week 3: Networking for hackers. The networking module of TryHackMe's Pre Security path; Wireshark: capture your own DNS lookup and TCP handshake. Proof: A screenshot of SYN, SYN-ACK, ACK with your explanation. 30 minutes a day and 2 hours on Sunday; a lab week can take two calendar weeks.
AWS	Module 4 (EC2 and EBS): instance types, AMIs, security groups. Start module 5 (RDS, DynamoDB).
TenantWall	Saturday, 3 hours: Diagrams and paper update. Redraw Fig. 1 with AWS icons, adding CloudTrail, CloudWatch and SNS; Draw the method flowchart; Add the Layer 5: Audit and alerting section; send the paper to the professor.

DAY BY DAY

– **Mon 7 Dec** Striver: step 5, step 6

Striver: step 5, step 6
- Count number of substrings (Medium)
- Longest palindromic substring (Medium)
- Sum of beauty of all substrings (Medium)
- Introduction to a linked list (Easy)
- Insert a node (Easy)
- Delete a node (Easy)

Core CS: Operating systems · CPU scheduling
- FCFS, SJF, SRTF, priority, round robin
- Answer 3 questions on it out loud

– **Tue 8 Dec** Striver: step 6

Striver: step 6
- Length of a linked list and searching (Easy)
- Insert into a doubly linked list (Easy)
- Delete a node in a doubly linked list (Easy)
- Reverse a doubly linked list (Easy)
- Delete all occurrences of a key in a DLL (Medium)
- Find pairs with a given sum in a DLL (Medium)

Core CS: Operating systems · CPU scheduling
- Waiting, turnaround and response time; Gantt charts
- Answer 3 questions on it out loud

– **Wed 9 Dec** Striver: step 6

Striver: step 6
- Remove duplicates from a sorted DLL (Medium)
- Middle of a linked list (Easy)
- Reverse a linked list (iterative) (Easy)
- Reverse a linked list (recursive) (Easy)
- Detect a loop (Easy)
- Find the starting point of the loop (Medium)

Core CS: Operating systems · CPU scheduling
- Starvation and aging
- Answer 3 questions on it out loud

– **Thu 10 Dec** Striver: step 6

Striver: step 6
- Length of the loop (Medium)
- Check if a linked list is a palindrome (Medium)
- Segregate odd and even nodes (Medium)
- Remove the Nth node from the back (Medium)
- Delete the middle node (Medium)
- Merge two sorted linked lists (Easy)

Core CS: Operating systems · CPU scheduling
- Choosing a time quantum for round robin
- Answer 3 questions on it out loud

– **Fri 11 Dec** Striver: step 6

Striver: step 6
- Sort a linked list (Medium)
- Sort a linked list of 0s, 1s and 2s (Medium)
- Intersection of two linked lists (Medium)
- Add 1 to a number in a linked list (Medium)
- Add two numbers in linked lists (Medium)
- Rotate a linked list (Medium)

Core CS: Operating systems · CPU scheduling
- FCFS, SJF, SRTF, priority, round robin
- Answer 3 questions on it out loud

– **Sat 12 Dec** Striver: step 7

Striver: step 7

- Generate all binary strings (Medium)
- Generate parentheses (Medium)
- Print all subsequences (Medium)
- Count good numbers (Medium)
- Reverse a stack using recursion (Medium)
- Count subsequences with sum K (Medium)

TenantWall, 3 h (in place of LeetCode)

- Redraw Fig. 1 with AWS icons, adding CloudTrail, CloudWatch and SNS
- Draw the method flowchart
- Add the Layer 5: Audit and alerting section; send the paper to the professor

Core CS: Operating systems · CPU scheduling

- Waiting, turnaround and response time; Gantt charts
- Answer 3 questions on it out loud

– Sun 13 Dec Sunday

Sunday

- Weekly contest, then upsolve one problem
- Written weekly review: solved, failed and why, what changes
- 2 hours of cyber
- Rest

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 5 14 Dec to 20 Dec 2026

READINESS CHECK 1

DSA	Step 7 Recursion and backtracking, Step 8 Bit manipulation, Step 9 Stacks and queues: 36 new problems, 6 a day, Monday to Saturday.
LeetCode	2 problems a day on this week's Striver topic, 35 minutes each. Sunday contest, then upsolve one.
Core CS	Operating systems: Synchronisation and deadlock. Race conditions, critical sections; Mutex vs semaphore (binary and counting); Producer-consumer, readers-writers, dining philosophers; Deadlock: the four Coffman conditions; Prevention, avoidance (Banker's algorithm), detection and recovery.
Cyber	Lab week 3: Networking for hackers. The networking module of TryHackMe's Pre Security path; Wireshark: capture your own DNS lookup and TCP handshake. Proof: A screenshot of SYN, SYN-ACK, ACK with your explanation. 30 minutes a day and 2 hours on Sunday; a lab week can take two calendar weeks.
AWS	Finish module 5, then module 6 (VPC): subnets, route tables, internet and NAT gateways.
College	Semester starts on 14 Dec. On college days (1 to 2 a week) do the light blocks: revision, Core CS, AWS and 2 Striver problems.

DAY BY DAY

– Mon 14 Dec Striver: step 7

Striver: step 7

- Check if a subsequence with sum K exists (Medium)
- Combination sum I (Medium)
- Combination sum II (Medium)
- Combination sum III (Medium)
- Subset sum I (Medium)
- Subsets II (with duplicates) (Medium)

Core CS: Operating systems · Synchronisation and deadlock

- Race conditions, critical sections
- Answer 3 questions on it out loud

– Tue 15 Dec Striver: step 7, step 8

Striver: step 7, step 8

- Letter combinations of a phone number (Medium)
- Print all permutations (Medium)
- Palindrome partitioning (Medium)
- Word search (Medium)
- Rat in a maze (Medium)

- Introduction to bit manipulation (AND, OR, XOR, shifts) (Easy)

Core CS: Operating systems · Synchronisation and deadlock

- Mutex vs semaphore (binary and counting)
- Answer 3 questions on it out loud

– **Wed 16 Dec** Striver: step 8

Striver: step 8

- Convert decimal to binary and back (Easy)
- 1's and 2's complement (Easy)
- Check if the i-th bit is set (Easy)
- Check if a number is odd or even (Easy)
- Check if a number is a power of 2 (Easy)
- Count the number of set bits (Easy)

Core CS: Operating systems · Synchronisation and deadlock

- Producer-consumer, readers-writers, dining philosophers
- Answer 3 questions on it out loud

– **Thu 17 Dec** Striver: step 8

Striver: step 8

- Set, clear and toggle the i-th bit (Easy)
- Remove the last set bit (Easy)
- Set the rightmost unset bit (Easy)
- Swap two numbers without a temporary (Easy)
- Divide two integers without * and / (Medium)
- Minimum bit flips to convert a number (Easy)

Core CS: Operating systems · Synchronisation and deadlock

- Deadlock: the four Coffman conditions
- Answer 3 questions on it out loud

– **Fri 18 Dec** Striver: step 8

Striver: step 8

- Single number I (Easy)
- Single number II (Medium)
- Single number III (Medium)
- XOR of numbers in a range L to R (Medium)
- Power set using bit manipulation (Medium)
- Print prime factors of a number (Medium)

Core CS: Operating systems · Synchronisation and deadlock

- Prevention, avoidance (Banker's algorithm), detection and recovery
- Answer 3 questions on it out loud

– **Sat 19 Dec** Striver: step 8, step 9

Striver: step 8, step 9

- Print all divisors of a number (Easy)
- Sieve of Eratosthenes (Medium)
- Prime factorisation using a sieve (Medium)
- Count primes in a range L to R (Medium)
- Power(x, n) (Medium)
- Implement a stack using arrays (Easy)

Core CS: Operating systems · Synchronisation and deadlock

- Race conditions, critical sections
- Answer 3 questions on it out loud

– **Sun 20 Dec** Sunday

Sunday

- Weekly contest, then upsolve one problem
- Written weekly review: solved, failed and why, what changes
- 2 hours of cyber
- Rest

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 6 21 Dec to 27 Dec 2026

DSA	Step 9 Stacks and queues, Step 10 Sliding window and two pointers, Step 11 Heaps: 36 new problems, 6 a day, Monday to Saturday.
LeetCode	2 problems a day on this week's Striver topic, 35 minutes each. Sunday contest, then upsolve one.
Core CS	Operating systems: Memory and files. Contiguous allocation, fragmentation; Paging, page tables, the TLB; segmentation; Virtual memory, demand paging, page faults; Page replacement: FIFO, LRU, Optimal; Belady's anomaly; thrashing; File systems basics: inodes, directories.
Cyber	Lab week 4: Scanning and bash. Bandit 16 to 20; TryHackMe's Nmap rooms; nmap your own home router only. Proof: Your nmap cheat sheet (template on the next pages). 30 minutes a day and 2 hours on Sunday; a lab week can take two calendar weeks.
AWS	Module 7 (connecting networks) and module 10 (CloudFormation).
TenantWall	Ask the professor for the venue and the deadline.
Backend project	Sunday, 4 hours. Design it: a study-tracker REST API. Write the endpoints and the SQL schema; create the repo.
Profile	Resume version 1. Set up GitHub (pinned repos) and LinkedIn.

DAY BY DAY

– Mon 21 Dec Striver: step 9

Striver: step 9

- Implement a queue using arrays (Easy)
- Implement a stack using queues (Easy)
- Implement a queue using stacks (Easy)
- Implement a stack using a linked list (Easy)
- Implement a queue using a linked list (Easy)
- Check for balanced parentheses (Easy)

Core CS: Operating systems · Memory and files

- Contiguous allocation, fragmentation
- Answer 3 questions on it out loud

– Tue 22 Dec Striver: step 9

Striver: step 9

- Implement a min stack (Medium)
- Infix to postfix (Medium)
- Prefix to infix (Medium)
- Prefix to postfix (Medium)
- Postfix to prefix (Medium)
- Postfix to infix (Medium)

Core CS: Operating systems · Memory and files

- Paging, page tables, the TLB; segmentation
- Answer 3 questions on it out loud

– Wed 23 Dec Striver: step 9

Striver: step 9

- Infix to prefix (Medium)
- Next greater element I (Easy)
- Next greater element II (circular) (Medium)
- Next smaller element (Medium)
- Sum of subarray minimums (Medium)
- Asteroid collision (Medium)

Core CS: Operating systems · Memory and files

- Virtual memory, demand paging, page faults
- Answer 3 questions on it out loud

– Thu 24 Dec Striver: step 9, step 10

Striver: step 9, step 10

- Sum of subarray ranges (Medium)
- Remove K digits (Medium)
- Stock span problem (Medium)

- The celebrity problem (Medium)
- Maximum points you can obtain from cards (Medium)
- Longest substring without repeating characters (Medium)

Core CS: Operating systems · Memory and files

- Page replacement: FIFO, LRU, Optimal; Belady's anomaly; thrashing
- Answer 3 questions on it out loud

– **Fri 25 Dec** Striver: step 10

Striver: step 10

- Max consecutive ones III (Medium)
- Fruit into baskets (Medium)
- Longest substring with at most K distinct characters (Medium)
- Longest repeating character replacement (Medium)
- Number of substrings containing all three characters (Medium)
- Binary subarrays with sum (Medium)

Core CS: Operating systems · Memory and files

- File systems basics: inodes, directories
- Answer 3 questions on it out loud

– **Sat 26 Dec** Striver: step 10, step 11

Striver: step 10, step 11

- Count number of nice subarrays (Medium)
- Introduction to a priority queue (Easy)
- Implement a min heap (Medium)
- Implement a max heap (Medium)
- Heapify and build a heap (Medium)
- Check if an array represents a min heap (Easy)

Core CS: Operating systems · Memory and files

- Contiguous allocation, fragmentation
- Answer 3 questions on it out loud

– **Sun 27 Dec** Sunday

Sunday

- Weekly contest, then upsolve one problem
- Written weekly review: solved, failed and why, what changes
- 2 hours of cyber; 4 hours on the backend project
- Rest

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 7 **28 Dec 2026 to 3 Jan 2027**

DSA	Step 11 Heaps, Step 12 Greedy algorithms, Step 13 Binary trees: 35 new problems, 6 a day, Monday to Saturday.
LeetCode	2 problems a day on this week's Striver topic, 35 minutes each. Sunday contest, then upsolve one.
Core CS	DBMS and SQL: Modelling. DBMS vs file system; ER model: entities, attributes, relationships, cardinality; Keys: super, candidate, primary, alternate, foreign, composite; Relational model and relational algebra basics.
Cyber	Lab week 5: HTTP and Burp Suite. Install Burp Suite Community; HTTP methods, headers, cookies, sessions; first PortSwigger Academy SQL injection labs. Proof: Burp history of one full login request, annotated. 30 minutes a day and 2 hours on Sunday; a lab week can take two calendar weeks.
AWS	Modules 11 and 12: caching (CloudFront, ElastiCache) and decoupling (SQS, SNS).
Backend project	Sunday, 4 hours. Build the CRUD endpoints with a real database (PostgreSQL or MySQL).
Internship	The first 5 applications to good companies. Ask the placement office what counts for the NOC.

DAY BY DAY

– **Mon 28 Dec** Striver: step 11

Striver: step 11

- Convert a min heap to a max heap (Medium)
- Heap sort (Medium)
- Kth largest element in an array (Medium)
- Kth smallest element (Medium)
- Sort a k-sorted array (Medium)
- Merge K sorted arrays (Medium)

Core CS: DBMS and SQL · Modelling

- DBMS vs file system
- Answer 3 questions on it out loud

– **Tue 29 Dec** Striver: step 11

Striver: step 11

- Replace elements by their rank (Medium)
- Task scheduler (Medium)
- Hand of straights (Medium)
- Design Twitter (Medium)
- Connect N ropes (Medium)
- Kth largest element in a stream (Medium)

Core CS: DBMS and SQL · Modelling

- ER model: entities, attributes, relationships, cardinality
- Answer 3 questions on it out loud

– **Wed 30 Dec** Striver: step 11, step 12

Striver: step 11, step 12

- Top K frequent elements (Medium)
- Assign cookies (Easy)
- Fractional knapsack (Medium)
- Minimum coins (greedy) (Easy)
- Lemonade change (Easy)
- Valid parenthesis string (Medium)

Core CS: DBMS and SQL · Modelling

- Keys: super, candidate, primary, alternate, foreign, composite
- Answer 3 questions on it out loud

– **Thu 31 Dec** Striver: step 12

Striver: step 12

- Jump game (Medium)
- Jump game II (Medium)
- Job sequencing problem (Medium)
- N meetings in one room (Easy)
- Minimum number of platforms (Medium)
- Shortest job first scheduling (Easy)

Core CS: DBMS and SQL · Modelling

- Relational model and relational algebra basics
- Answer 3 questions on it out loud

– **Fri 1 Jan** Striver: step 12, step 13

Striver: step 12, step 13

- Page faults in LRU (Easy)
- Non-overlapping intervals (Medium)
- Insert interval (Medium)
- Merge intervals (revisit) (Medium)
- Introduction to trees and representation (Easy)
- Preorder traversal (Easy)

Core CS: DBMS and SQL · Modelling

- DBMS vs file system
- Answer 3 questions on it out loud

– **Sat 2 Jan** Striver: step 13

Striver: step 13

- Inorder traversal (Easy)
- Postorder traversal (Easy)
- Level order traversal (Medium)

- Iterative preorder (Medium)
- Iterative inorder (Medium)

Core CS: DBMS and SQL · Modelling

- ER model: entities, attributes, relationships, cardinality
- Answer 3 questions on it out loud

– Sun 3 Jan Sunday

Sunday

- Weekly contest, then upsolve one problem
- Written weekly review: solved, failed and why, what changes
- 2 hours of cyber; 4 hours on the backend project
- Rest

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 8 4 Jan to 10 Jan 2027

DSA	Step 13 Binary trees, Step 14 Binary search trees: 30 new problems, 5 a day, Monday to Saturday.
LeetCode	2 problems a day on this week's Striver topic, 35 minutes each. Sunday contest, then upsolve one.
Core CS	DBMS and SQL: SQL. SELECT, WHERE, ORDER BY, GROUP BY, HAVING; Joins: inner, left, right, full, self, cross; Subqueries and common table expressions; Window functions: ROW_NUMBER, RANK, DENSE_RANK; Start LeetCode's SQL 50 study plan: about 7 a week.
Cyber	Lab week 5: HTTP and Burp Suite. Install Burp Suite Community; HTTP methods, headers, cookies, sessions; first PortSwigger Academy SQL injection labs. Proof: Burp history of one full login request, annotated. 30 minutes a day and 2 hours on Sunday; a lab week can take two calendar weeks.
AWS	Modules 13 and 14: serverless (Lambda, API Gateway) and planning for disaster.
Backend project	Sunday, 4 hours. Add sign-up and login with hashed passwords and JWT; protect the routes.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.

DAY BY DAY

– Mon 4 Jan Striver: step 13

Striver: step 13

- Iterative postorder (two stacks) (Medium)
- Iterative postorder (one stack) (Medium)
- Maximum depth of a binary tree (Easy)
- Check for a balanced binary tree (Medium)
- Diameter of a binary tree (Medium)

Core CS: DBMS and SQL · SQL

- SELECT, WHERE, ORDER BY, GROUP BY, HAVING
- Answer 3 questions on it out loud

– Tue 5 Jan Striver: step 13

Striver: step 13

- Check if two trees are identical (Easy)
- Zigzag traversal (Medium)
- Boundary traversal (Medium)
- Top view (Medium)
- Bottom view (Medium)

Core CS: DBMS and SQL · SQL

- Joins: inner, left, right, full, self, cross
- Answer 3 questions on it out loud

– Wed 6 Jan Striver: step 13

Striver: step 13

- Right and left view (Medium)

- Symmetric binary tree (Easy)
- Root to node path (Medium)
- Lowest common ancestor (Medium)
- Maximum width of a binary tree (Medium)

Core CS: DBMS and SQL · SQL

- Subqueries and common table expressions
- Answer 3 questions on it out loud

– Thu 7 Jan Striver: step 13

Striver: step 13

- Children sum property (Medium)
- Count nodes in a complete binary tree (Medium)
- Construct a tree from preorder and inorder (Medium)
- Construct a tree from postorder and inorder (Medium)
- Flatten a binary tree to a linked list (Medium)

Core CS: DBMS and SQL · SQL

- Window functions: ROW_NUMBER, RANK, DENSE_RANK
- Answer 3 questions on it out loud

– Fri 8 Jan Striver: step 14

Striver: step 14

- Introduction to a BST (Easy)
- Search in a BST (Easy)
- Minimum and maximum in a BST (Easy)
- Ceil in a BST (Medium)
- Floor in a BST (Medium)

Core CS: DBMS and SQL · SQL

- Start LeetCode's SQL 50 study plan: about 7 a week
- Answer 3 questions on it out loud

– Sat 9 Jan Striver: step 14

Striver: step 14

- Insert a node (Medium)
- Kth smallest element (Medium)
- Kth largest element (Medium)
- Check if a tree is a BST (Medium)
- Lowest common ancestor in a BST (Medium)

Core CS: DBMS and SQL · SQL

- SELECT, WHERE, ORDER BY, GROUP BY, HAVING
- Answer 3 questions on it out loud

– Sun 10 Jan Sunday

Sunday

- Weekly contest, then upsolve one problem
- Written weekly review: solved, failed and why, what changes
- 2 hours of cyber; 4 hours on the backend project
- Rest

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 9 11 Jan to 17 Jan 2027

AWS COURSE AND PRACTICE EXAM DONE

DSA	Step 14 Binary search trees, Step 15 Graphs: 30 new problems, 5 a day, Monday to Saturday.
LeetCode	2 problems a day on this week's Striver topic, 35 minutes each. Sunday contest, then upsolve one.
Core CS	DBMS and SQL: Normalisation. Functional dependencies; Insert, update and delete anomalies; 1NF, 2NF, 3NF, BCNF with examples; When to denormalise.
Cyber	Lab week 6: SQL injection. Remaining apprentice SQLi labs including UNION attacks; picoCTF web exploitation challenges. Proof: Writeup: one SQLi lab, payload by payload. 30 minutes a day and 2 hours on Sunday; a lab week can take two calendar weeks.
AWS	Module 15 and its practice exam.

TenantWall	Start the GitHub repo: policies, scripts, diagrams, README.
Backend project	Sunday, 4 hours. Write tests for every endpoint; add input validation and error handling.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.

DAY BY DAY

– **Mon 11 Jan** Striver: step 14, step 15

Striver: step 14, step 15
- Construct a BST from preorder (Medium)
- Inorder successor and predecessor (Medium)
- BST iterator (Medium)
- Two sum IV in a BST (Medium)
- Graph representation (matrix and list) (Easy)

Core CS: DBMS and SQL · Normalisation
- Functional dependencies
- Answer 3 questions on it out loud

– **Tue 12 Jan** Striver: step 15

Striver: step 15
- BFS traversal (Easy)
- DFS traversal (Easy)
- Number of provinces (Medium)
- Connected components (Medium)
- Number of islands (Medium)

Core CS: DBMS and SQL · Normalisation
- Insert, update and delete anomalies
- Answer 3 questions on it out loud

– **Wed 13 Jan** Striver: step 15

Striver: step 15
- Flood fill (Medium)
- Rotten oranges (Medium)
- Detect a cycle in an undirected graph (BFS) (Medium)
- Detect a cycle in an undirected graph (DFS) (Medium)
- Distance of the nearest 1 in a 0/1 matrix (Medium)

Core CS: DBMS and SQL · Normalisation
- 1NF, 2NF, 3NF, BCNF with examples
- Answer 3 questions on it out loud

– **Thu 14 Jan** Striver: step 15

Striver: step 15
- Surrounded regions (Medium)
- Number of enclaves (Medium)
- Number of distinct islands (Medium)
- Bipartite graph (Medium)
- Detect a cycle in a directed graph (DFS) (Medium)

Core CS: DBMS and SQL · Normalisation
- When to denormalise
- Answer 3 questions on it out loud

– **Fri 15 Jan** Striver: step 15

Striver: step 15
- Find eventual safe states (Medium)
- Topological sort (DFS) (Medium)
- Kahn's algorithm (BFS) (Medium)
- Cycle detection in a directed graph (BFS) (Medium)
- Course schedule I (Medium)

Core CS: DBMS and SQL · Normalisation
- Functional dependencies
- Answer 3 questions on it out loud

– Sat 16 Jan Striver: step 15

Striver: step 15

- Course schedule II (Medium)
- Shortest path in an undirected graph with unit weights (Medium)
- Shortest path in a DAG (Medium)
- Dijkstra's algorithm (Medium)
- Shortest path in a binary maze (Medium)

Core CS: DBMS and SQL · Normalisation

- Insert, update and delete anomalies
- Answer 3 questions on it out loud

– Sun 17 Jan Sunday

Sunday

- Weekly contest, then upsolve one problem
- Written weekly review: solved, failed and why, what changes
- 2 hours of cyber; 4 hours on the backend project
- Rest

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 10 18 Jan to 24 Jan 2027

DSA	Step 15 Graphs, Step 16 Dynamic programming: 30 new problems, 5 a day, Monday to Saturday.
LeetCode	2 problems a day on this week's Striver topic, 35 minutes each. Sunday contest, then upsolve one.
Core CS	DBMS and SQL: Transactions and indexes. ACID properties; Isolation levels and the anomalies they prevent; Locks and two-phase locking; Indexes: B+ trees, clustered vs non-clustered; SQL vs NoSQL; CAP theorem in one paragraph.
Cyber	Lab week 7: Cross-site scripting. PortSwigger reflected, stored and DOM XSS apprentice labs. Proof: A table: XSS type, where it lands, how to fix it. 30 minutes a day and 2 hours on Sunday; a lab week can take two calendar weeks.
AWS	Labs from memory: a VPC with public and private subnets and a NAT gateway; an IAM role for EC2.
Backend project	Sunday, 4 hours. Deploy it on AWS (EC2 with RDS, free tier); add a health check.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.
Mock interviews	Your first mock interview: one DSA problem, thinking aloud, with a classmate or senior.

DAY BY DAY

– Mon 18 Jan Striver: step 15

Striver: step 15

- Path with minimum effort (Medium)
- Cheapest flights within K stops (Medium)
- Network delay time (Medium)
- Minimum multiplications to reach the end (Medium)
- Number of ways to arrive at the destination (Medium)

Core CS: DBMS and SQL · Transactions and indexes

- ACID properties
- Answer 3 questions on it out loud

– Tue 19 Jan Striver: step 15

Striver: step 15

- Bellman-Ford algorithm (Medium)
- Floyd-Warshall algorithm (Medium)
- City with the smallest number of neighbours at a threshold (Medium)
- Minimum spanning tree: Prim's algorithm (Medium)
- Disjoint set (rank and path compression) (Medium)

Core CS: DBMS and SQL · Transactions and indexes

- Isolation levels and the anomalies they prevent
- Answer 3 questions on it out loud

– Wed 20 Jan Striver: step 15, step 16

Striver: step 15, step 16
- Kruskal's algorithm (Medium)
- Number of operations to make a network connected (Medium)
- Most stones removed (Medium)
- Accounts merge (Medium)
- Introduction to DP (Fibonacci with memoisation) (Easy)

Core CS: DBMS and SQL · Transactions and indexes
- Locks and two-phase locking
- Answer 3 questions on it out loud

– Thu 21 Jan Striver: step 16

Striver: step 16
- Climbing stairs (Easy)
- Frog jump (Easy)
- Frog jump with K distances (Medium)
- Maximum sum of non-adjacent elements (Medium)
- House robber II (Medium)

Core CS: DBMS and SQL · Transactions and indexes
- Indexes: B+ trees, clustered vs non-clustered
- Answer 3 questions on it out loud

– Fri 22 Jan Striver: step 16

Striver: step 16
- Ninja's training (Medium)
- Grid unique paths (Medium)
- Unique paths II (obstacles) (Medium)
- Minimum path sum in a grid (Medium)
- Triangle minimum path sum (Medium)

Core CS: DBMS and SQL · Transactions and indexes
- SQL vs NoSQL; CAP theorem in one paragraph
- Answer 3 questions on it out loud

– Sat 23 Jan Striver: step 16

Striver: step 16
- Minimum and maximum falling path sum (Medium)
- Subset sum equal to target (Medium)
- Partition equal subset sum (Medium)
- Partition into two subsets with minimum absolute difference (Medium)
- Count subsets with sum K (Medium)

Core CS: DBMS and SQL · Transactions and indexes
- ACID properties
- Answer 3 questions on it out loud

– Sun 24 Jan Sunday

Sunday
- Weekly contest, then upsolve one problem
- Written weekly review: solved, failed and why, what changes
- 2 hours of cyber; 4 hours on the backend project
- Rest

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 11 25 Jan to 31 Jan 2027 STRIVER CORE DONE

DSA	Step 16 Dynamic programming, Step 17 Tries, Step 18 Strings (advanced): 30 new problems, 5 a day, Monday to Saturday. Striver core ends this week; then redo every problem marked H or F.
LeetCode	2 problems a day on this week's Striver topic, 35 minutes each. Sunday contest, then upsolve one.

Core CS	Computer networks: Models and devices. OSI 7 layers vs TCP/IP 4 layers; encapsulation; Hub, switch, router, gateway; collision and broadcast domains; MAC vs IP addresses; ARP; Ethernet frames basics.
Cyber	Lab week 7: Cross-site scripting. PortSwigger reflected, stored and DOM XSS apprentice labs. Proof: A table: XSS type, where it lands, how to fix it. 30 minutes a day and 2 hours on Sunday; a lab week can take two calendar weeks.
AWS	Labs from memory: EC2 behind a load balancer with Auto Scaling; RDS in a private subnet.
Backend project	Sunday, 4 hours. README, architecture diagram, and the resume line with one real number.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.

DAY BY DAY

– **Mon 25 Jan** Striver: step 16

Striver: step 16
- Count partitions with a given difference (Medium)
- 0/1 knapsack (Medium)
- Minimum coins (Medium)
- Target sum (Medium)
- Coin change II (Medium)

Core CS: Computer networks · Models and devices
- OSI 7 layers vs TCP/IP 4 layers; encapsulation
- Answer 3 questions on it out loud

– **Tue 26 Jan** Striver: step 16

Striver: step 16
- Unbounded knapsack (Medium)
- Rod cutting (Medium)
- Longest common subsequence (Medium)
- Print the LCS (Medium)
- Longest common substring (Medium)

Core CS: Computer networks · Models and devices
- Hub, switch, router, gateway; collision and broadcast domains
- Answer 3 questions on it out loud

– **Wed 27 Jan** Striver: step 16

Striver: step 16
- Longest palindromic subsequence (Medium)
- Minimum insertions to make a string a palindrome (Medium)
- Minimum insertions and deletions to convert a string (Medium)
- Best time to buy and sell stock II (Medium)
- Stock with cooldown (Medium)

Core CS: Computer networks · Models and devices
- MAC vs IP addresses; ARP
- Answer 3 questions on it out loud

– **Thu 28 Jan** Striver: step 16

Striver: step 16
- Stock with transaction fee (Medium)
- Longest increasing subsequence (Medium)
- Print the LIS (Medium)
- Largest divisible subset (Medium)
- Longest string chain (Medium)

Core CS: Computer networks · Models and devices
- Ethernet frames basics
- Answer 3 questions on it out loud

– **Fri 29 Jan** Striver: step 16, step 17

Striver: step 16, step 17
- Longest bitonic subsequence (Medium)
- Number of longest increasing subsequences (Medium)

- Partition array for maximum sum (Medium)
- Count square submatrices with all 1s (Medium)
- Implement a trie (prefix tree) I (Medium)

Core CS: Computer networks · Models and devices

- OSI 7 layers vs TCP/IP 4 layers; encapsulation
- Answer 3 questions on it out loud

– Sat 30 Jan Striver: step 17, step 18

Striver: step 17, step 18

- Implement a trie II (Medium)
- Longest word with all prefixes (Medium)
- Number of distinct substrings in a string using a trie (Medium)
- Count and say (Medium)
- Minimum bracket reversals to balance an expression (Medium)

Core CS: Computer networks · Models and devices

- Hub, switch, router, gateway; collision and broadcast domains
- Answer 3 questions on it out loud

– Sun 31 Jan Sunday

Sunday

- Weekly contest, then upsolve one problem
- Written weekly review: solved, failed and why, what changes
- 2 hours of cyber; 4 hours on the backend project
- Rest

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 12 1 Feb to 7 Feb 2027

DSA	Redo every problem marked H or F from memory, then mixed timed sets: 3 problems in 90 minutes.
LeetCode	2 problems a day on this week's Striver topic, 35 minutes each. Sunday contest, then upsolve one.
Core CS	Computer networks: IP and routing. IPv4 addressing, subnetting and CIDR; Private ranges, NAT, DHCP; ICMP, ping and traceroute; Routing basics; IPv6 at a glance.
Cyber	Lab week 8: Access control and auth. PortSwigger access control (IDOR) and authentication apprentice labs. Proof: Writeup: one IDOR, with the fix in code. 30 minutes a day and 2 hours on Sunday; a lab week can take two calendar weeks.
AWS	Labs from memory: S3 with CloudFront; SQS plus Lambda; a CloudFormation template that rebuilds one lab.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Profile	Resume version 2, with the backend project and your problem counts.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.
Mock interviews	Two mock interviews and two timed online-test sets.

DAY BY DAY

– Mon 1 Feb Revision

Revision

- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

Core CS: Computer networks · IP and routing

- IPv4 addressing, subnetting and CIDR
- Answer 3 questions on it out loud

– Tue 2 Feb Revision

Revision

- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

Core CS: Computer networks · IP and routing

- Private ranges, NAT, DHCP
- Answer 3 questions on it out loud

– **Wed 3 Feb** Revision

Revision
- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

Core CS: Computer networks · IP and routing
- ICMP, ping and traceroute
- Answer 3 questions on it out loud

– **Thu 4 Feb** Revision

Revision
- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

Core CS: Computer networks · IP and routing
- Routing basics; IPv6 at a glance
- Answer 3 questions on it out loud

– **Fri 5 Feb** Revision

Revision
- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

Core CS: Computer networks · IP and routing
- IPv4 addressing, subnetting and CIDR
- Answer 3 questions on it out loud

– **Sat 6 Feb** Revision

Revision
- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

Core CS: Computer networks · IP and routing
- Private ranges, NAT, DHCP
- Answer 3 questions on it out loud

– **Sun 7 Feb** Sunday

Sunday
- Weekly contest, then upsolve one problem
- Written weekly review: solved, failed and why, what changes
- 2 hours of cyber
- Rest

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

Build

– WEEK 13 **8 Feb to 14 Feb 2027**

DSA	Hard pass: 7 Hard problems, one a day, from Arrays. 30 minutes alone before any hint.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 800: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	Computer networks: Transport. TCP vs UDP; ports; Three-way handshake and four-way close; Flow control (sliding window) vs congestion control (slow start, AIMD); Reliability: sequence numbers, ACKs, retransmission.
Cyber	Lab week 9: OWASP Top 10 review. Read the OWASP Top 10; PortSwigger CSRF and SSRF apprentice labs. Proof: Your OWASP map: each item linked to a lab you did. One hour a day.
AWS	Tie it to your profile: TenantWall write-up with an AWS-style diagram, one AWS project on GitHub, a resume line.
AI	Week 1 of 12: How LLMs work, and your first API call. Andrej Karpathy's one-hour talk "Intro to Large Language Models" and 3Blue1Brown's videos on GPT and attention. Then get a free API key (for example Google AI Studio) or run a local model with Ollama, call it from a Python script, and keep the key in an environment variable. Proof: One page on how LLMs work in your own words, and a script that summarises any text file with 3 temperature settings compared. Two hours a day.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.

Internship	5 applications to good companies and one referral request. Stop when you have an offer.
Mock interviews	One mock interview. Note every question you could not answer.

DAY BY DAY

– **Mon 8 Feb** Hard pass

Hard pass
- Step 3: Pascal's triangle

Core CS: Computer networks · Transport
- TCP vs UDP; ports
- Answer 3 questions on it out loud

AI week 1: How LLMs work, and your first API call
- 2 hours toward: One page on how LLMs work in your own words, and a script that summarises any text file with 3 temperature settings compared

– **Tue 9 Feb** Hard pass

Hard pass
- Step 3: Majority element (more than n/3)

Core CS: Computer networks · Transport
- Three-way handshake and four-way close
- Answer 3 questions on it out loud

AI week 1: How LLMs work, and your first API call
- 2 hours toward: One page on how LLMs work in your own words, and a script that summarises any text file with 3 temperature settings compared

– **Wed 10 Feb** Hard pass

Hard pass
- Step 3: 3-sum

Core CS: Computer networks · Transport
- Flow control (sliding window) vs congestion control (slow start, AIMD)
- Answer 3 questions on it out loud

AI week 1: How LLMs work, and your first API call
- 2 hours toward: One page on how LLMs work in your own words, and a script that summarises any text file with 3 temperature settings compared

– **Thu 11 Feb** Hard pass

Hard pass
- Step 3: 4-sum

Core CS: Computer networks · Transport
- Reliability: sequence numbers, ACKs, retransmission
- Answer 3 questions on it out loud

AI week 1: How LLMs work, and your first API call
- 2 hours toward: One page on how LLMs work in your own words, and a script that summarises any text file with 3 temperature settings compared

– **Fri 12 Feb** Hard pass

Hard pass
- Step 3: Largest subarray with sum 0

Core CS: Computer networks · Transport
- TCP vs UDP; ports
- Answer 3 questions on it out loud

AI week 1: How LLMs work, and your first API call
- 2 hours toward: One page on how LLMs work in your own words, and a script that summarises any text file with 3 temperature settings compared

– **Sat 13 Feb** Hard pass

Hard pass
- Step 3: Count subarrays with XOR K

Core CS: Computer networks · Transport
- Three-way handshake and four-way close
- Answer 3 questions on it out loud

AI week 1: How LLMs work, and your first API call

- 2 hours toward: One page on how LLMs work in your own words, and a script that summarises any text file with 3 temperature settings compared

– Sun 14 Feb Hard pass

Hard pass

- Step 3: Merge overlapping intervals

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 14 15 Feb to 21 Feb 2027

DSA	Hard pass: 7 Hard problems, one a day, from Arrays, Binary search. 30 minutes alone before any hint.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 800: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	Computer networks: Application. DNS resolution and record types; HTTP methods, status codes, headers, cookies; HTTPS and the TLS handshake; What happens when you type a URL, end to end.
Cyber	Lab week 10: Cryptography basics. Hashing vs encryption, salts, symmetric vs asymmetric, certificates, JWTs; picoCTF cryptography challenges (5). Proof: One-page crypto cheat sheet in your own words. One hour a day.
AWS	Exam prep: read the SAA-C03 exam guide; start the secure-architectures domain (30%). 10 practice questions a day from now on.
AI	Week 2 of 12: Prompting and structured output. System prompts, examples in the prompt (few-shot), asking for JSON and validating it. Read one model provider's prompt-engineering guide. Proof: A prompt that turns a job post into JSON (role, skills, stipend), tested on 10 real posts, with its pass rate. Two hours a day.
TenantWall	Draft the final paper with the audit and alerting section.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.
Mock interviews	One mock interview. Note every question you could not answer.

DAY BY DAY

– Mon 15 Feb Hard pass

Hard pass

- Step 3: Merge two sorted arrays without extra space

Core CS: Computer networks · Application

- DNS resolution and record types
- Answer 3 questions on it out loud

AI week 2: Prompting and structured output

- 2 hours toward: A prompt that turns a job post into JSON (role, skills, stipend), tested on 10 real posts, with its pass rate

– Tue 16 Feb Hard pass

Hard pass

- Step 3: Find the repeating and missing number

Core CS: Computer networks · Application

- HTTP methods, status codes, headers, cookies
- Answer 3 questions on it out loud

AI week 2: Prompting and structured output

- 2 hours toward: A prompt that turns a job post into JSON (role, skills, stipend), tested on 10 real posts, with its pass rate

– Wed 17 Feb Hard pass

Hard pass

- Step 3: Count inversions

Core CS: Computer networks · Application

- HTTPS and the TLS handshake
- Answer 3 questions on it out loud

AI week 2: Prompting and structured output

- 2 hours toward: A prompt that turns a job post into JSON (role, skills, stipend), tested on 10 real posts, with its pass rate

– Thu 18 Feb Hard pass

Hard pass

- Step 3: Reverse pairs

Core CS: Computer networks · Application

- What happens when you type a URL, end to end
- Answer 3 questions on it out loud

AI week 2: Prompting and structured output

- 2 hours toward: A prompt that turns a job post into JSON (role, skills, stipend), tested on 10 real posts, with its pass rate

– Fri 19 Feb Hard pass

Hard pass

- Step 3: Maximum product subarray

Core CS: Computer networks · Application

- DNS resolution and record types
- Answer 3 questions on it out loud

AI week 2: Prompting and structured output

- 2 hours toward: A prompt that turns a job post into JSON (role, skills, stipend), tested on 10 real posts, with its pass rate

– Sat 20 Feb Hard pass

Hard pass

- Step 4: Aggressive cows

Core CS: Computer networks · Application

- HTTP methods, status codes, headers, cookies
- Answer 3 questions on it out loud

AI week 2: Prompting and structured output

- 2 hours toward: A prompt that turns a job post into JSON (role, skills, stipend), tested on 10 real posts, with its pass rate

– Sun 21 Feb Hard pass

Hard pass

- Step 4: Book allocation

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 15 22 Feb to 28 Feb 2027

DSA	Hard pass: 7 Hard problems, one a day, from Binary search. 30 minutes alone before any hint.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 900: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	System design basics: Scaling a web service. Client-server, latency vs throughput; Vertical vs horizontal scaling; stateless services; Load balancers and their algorithms; Caching: where, what, eviction (LRU), TTL; CDNs.
Cyber	Lab week 11: Packet analysis. Wireshark display filters (dns, http, tcp.flags.syn == 1), Follow TCP Stream; picoCTF forensics pcap challenges. Pairs with your CN month.. Proof: Writeup: one pcap challenge. One hour a day.
AWS	Secure architectures, continued: KMS, VPC security, S3 security. Map AWS security services to your cyber skills.
AI	Week 3 of 12: Embeddings and search. Embeddings and cosine similarity; splitting text into chunks; a free local embedding model (sentence-transformers) and a vector store (Chroma or FAISS). Proof: Search over your own Core CS notes: 5 queries and what came back. Two hours a day.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.

Internship	5 applications to good companies and one referral request. Stop when you have an offer.
Mock interviews	One mock interview. Note every question you could not answer.

DAY BY DAY

– **Mon 22 Feb** Hard pass

Hard pass
- Step 4: Split array largest sum

Core CS: System design basics · Scaling a web service
- Client-server, latency vs throughput
- Answer 3 questions on it out loud

AI week 3: Embeddings and search
- 2 hours toward: Search over your own Core CS notes: 5 queries and what came back

– **Tue 23 Feb** Hard pass

Hard pass
- Step 4: Painter's partition

Core CS: System design basics · Scaling a web service
- Vertical vs horizontal scaling; stateless services
- Answer 3 questions on it out loud

AI week 3: Embeddings and search
- 2 hours toward: Search over your own Core CS notes: 5 queries and what came back

– **Wed 24 Feb** Hard pass

Hard pass
- Step 4: Minimise maximum distance to a gas station

Core CS: System design basics · Scaling a web service
- Load balancers and their algorithms
- Answer 3 questions on it out loud

AI week 3: Embeddings and search
- 2 hours toward: Search over your own Core CS notes: 5 queries and what came back

– **Thu 25 Feb** Hard pass

Hard pass
- Step 4: Median of two sorted arrays

Core CS: System design basics · Scaling a web service
- Caching: where, what, eviction (LRU), TTL; CDNs
- Answer 3 questions on it out loud

AI week 3: Embeddings and search
- 2 hours toward: Search over your own Core CS notes: 5 queries and what came back

– **Fri 26 Feb** Hard pass

Hard pass
- Step 4: Kth element of two sorted arrays

Core CS: System design basics · Scaling a web service
- Client-server, latency vs throughput
- Answer 3 questions on it out loud

AI week 3: Embeddings and search
- 2 hours toward: Search over your own Core CS notes: 5 queries and what came back

– **Sat 27 Feb** Hard pass

Hard pass
- Step 4: Find a peak element II

Core CS: System design basics · Scaling a web service
- Vertical vs horizontal scaling; stateless services
- Answer 3 questions on it out loud

AI week 3: Embeddings and search
- 2 hours toward: Search over your own Core CS notes: 5 queries and what came back

– **Sun 28 Feb** Hard pass

Hard pass

- Step 4: Median of a row-wise sorted matrix

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 16 1 Mar to 7 Mar 2027

DSA	Hard pass: 7 Hard problems, one a day, from Linked list, Recursion and backtracking. 30 minutes alone before any hint.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 900: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	System design basics: Data at scale. Replication (leader-follower) and read replicas; Sharding and consistent hashing; CAP theorem; consistency choices; Message queues and async work; rate limiting.
Cyber	Lab week 12: First machines. Hack The Box Starting Point, Tier 0 machines. Proof: Personal notes for each machine (keep them private). One hour a day.
AWS	Resilient architectures (26%): multi-AZ, Route 53, SQS, backups, disaster recovery.
AI	Week 4 of 12: RAG: ask your notes. Retrieve the best chunks, put them in the prompt, answer with citations, and say "not in my notes" when nothing matches. Proof: An "Ask my notes" repo on GitHub with a README and 5 example questions. Two hours a day.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.
Mock interviews	One mock interview. Note every question you could not answer.

DAY BY DAY

– Mon 1 Mar Hard pass

Hard pass

- Step 6: Reverse nodes in k-group

Core CS: System design basics · Data at scale

- Replication (leader-follower) and read replicas
- Answer 3 questions on it out loud

AI week 4: RAG: ask your notes

- 2 hours toward: An "Ask my notes" repo on GitHub with a README and 5 example questions

– Tue 2 Mar Hard pass

Hard pass

- Step 6: Flatten a linked list

Core CS: System design basics · Data at scale

- Sharding and consistent hashing
- Answer 3 questions on it out loud

AI week 4: RAG: ask your notes

- 2 hours toward: An "Ask my notes" repo on GitHub with a README and 5 example questions

– Wed 3 Mar Hard pass

Hard pass

- Step 6: Clone a linked list with random pointers

Core CS: System design basics · Data at scale

- CAP theorem; consistency choices
- Answer 3 questions on it out loud

AI week 4: RAG: ask your notes

- 2 hours toward: An "Ask my notes" repo on GitHub with a README and 5 example questions

– Thu 4 Mar Hard pass

Hard pass

- Step 7: Kth permutation sequence

Core CS: System design basics · Data at scale

- Message queues and async work; rate limiting
- Answer 3 questions on it out loud

AI week 4: RAG: ask your notes

- 2 hours toward: An "Ask my notes" repo on GitHub with a README and 5 example questions

– **Fri 5 Mar** Hard pass

Hard pass

- Step 7: N-Queens

Core CS: System design basics · Data at scale

- Replication (leader-follower) and read replicas
- Answer 3 questions on it out loud

AI week 4: RAG: ask your notes

- 2 hours toward: An "Ask my notes" repo on GitHub with a README and 5 example questions

– **Sat 6 Mar** Hard pass

Hard pass

- Step 7: Sudoku solver

Core CS: System design basics · Data at scale

- Sharding and consistent hashing
- Answer 3 questions on it out loud

AI week 4: RAG: ask your notes

- 2 hours toward: An "Ask my notes" repo on GitHub with a README and 5 example questions

– **Sun 7 Mar** Hard pass

Hard pass

- Step 7: M-colouring problem

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 17 **8 Mar to 14 Mar 2027**

DSA	Hard pass: 7 Hard problems, one a day, from Recursion and backtracking, Stacks and queues. 30 minutes alone before any hint.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 1000: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	System design basics: Design practice. Low-level design: parking lot, LRU cache, library system as classes; High-level design: URL shortener, then a chat service; Write TenantWall's design document with a diagram; Practise talking through a design for 20 minutes.
Cyber	Lab week 13: Privilege escalation. Starting Point Tier 1; TryHackMe Linux privilege escalation room: sudo -l, SUID, cron, PATH. Proof: PrivEsc checklist you can reuse. One hour a day.
AWS	High-performing architectures (24%): caching, EBS types, scaling, databases.
AI	Week 5 of 12: Tool calling and a small agent. Let the model call your own functions (search notes, a calculator, read a file). Log every step it takes. Proof: A command-line assistant with 2 tools and a printed trace of one run. Two hours a day.
TenantWall	Final read-through with the team.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.
Mock interviews	One mock interview. Note every question you could not answer.

DAY BY DAY

– **Mon 8 Mar** Hard pass

Hard pass

- Step 7: Word break

Core CS: System design basics · Design practice

- Low-level design: parking lot, LRU cache, library system as classes
- Answer 3 questions on it out loud

AI week 5: Tool calling and a small agent

- 2 hours toward: A command-line assistant with 2 tools and a printed trace of one run

– **Tue 9 Mar** Hard pass

Hard pass
- Step 7: Expression add operators

Core CS: System design basics · Design practice
- High-level design: URL shortener, then a chat service
- Answer 3 questions on it out loud

AI week 5: Tool calling and a small agent
- 2 hours toward: A command-line assistant with 2 tools and a printed trace of one run

– **Wed 10 Mar** Hard pass

Hard pass
- Step 9: Number of NGEs to the right

Core CS: System design basics · Design practice
- Write TenantWall's design document with a diagram
- Answer 3 questions on it out loud

AI week 5: Tool calling and a small agent
- 2 hours toward: A command-line assistant with 2 tools and a printed trace of one run

– **Thu 11 Mar** Hard pass

Hard pass
- Step 9: Trapping rain water

Core CS: System design basics · Design practice
- Practise talking through a design for 20 minutes
- Answer 3 questions on it out loud

AI week 5: Tool calling and a small agent
- 2 hours toward: A command-line assistant with 2 tools and a printed trace of one run

– **Fri 12 Mar** Hard pass

Hard pass
- Step 9: Largest rectangle in a histogram

Core CS: System design basics · Design practice
- Low-level design: parking lot, LRU cache, library system as classes
- Answer 3 questions on it out loud

AI week 5: Tool calling and a small agent
- 2 hours toward: A command-line assistant with 2 tools and a printed trace of one run

– **Sat 13 Mar** Hard pass

Hard pass
- Step 9: Maximal rectangles

Core CS: System design basics · Design practice
- High-level design: URL shortener, then a chat service
- Answer 3 questions on it out loud

AI week 5: Tool calling and a small agent
- 2 hours toward: A command-line assistant with 2 tools and a printed trace of one run

– **Sun 14 Mar** Hard pass

Hard pass
- Step 9: Sliding window maximum

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 18 **15 Mar to 21 Mar 2027** READINESS CHECK 2 PAPER SUBMITTED

DSA	Hard pass: 7 Hard problems, one a day, from Stacks and queues, Sliding window and two pointers, Heaps, Greedy algorithms. 30 minutes alone before any hint.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 1000: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	Revision: all five subjects in rotation, 5 questions a day out loud.
Cyber	Lab week 14: Logs. Linux auth logs, web server logs, Windows event IDs 4624, 4625, 4688, 4720; run your Step 10 logwatch on a sample log. Proof: Five detection rules written in plain

	English. One hour a day.
AWS	Cost-optimised architectures (20%): pricing models, storage classes, right-sizing. First full timed practice exam.
AI	Week 6 of 12: Ship an AI feature and test it. Add one AI endpoint to your web backend project: input limits, timeouts, error handling, the key in an environment variable, cost and latency logged. Then write 20 test questions with expected answers, measure accuracy and fix the worst 3 failures. Proof: A working endpoint, a before-and-after score table, and a resume line with one real number. Two hours a day.
TenantWall	Submit by Sunday 21 Mar, before exams.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.
Mock interviews	One mock interview. Note every question you could not answer.
College	Exams start next week. From Wednesday, give the Core CS and cyber time to exam subjects.

DAY BY DAY

– **Mon 15 Mar** Hard pass

Hard pass
- Step 9: LRU cache

AI week 6: Ship an AI feature and test it
- 2 hours toward: A working endpoint, a before-and-after score table, and a resume line with one real number

– **Tue 16 Mar** Hard pass

Hard pass
- Step 9: LFU cache

AI week 6: Ship an AI feature and test it
- 2 hours toward: A working endpoint, a before-and-after score table, and a resume line with one real number

– **Wed 17 Mar** Hard pass

Hard pass
- Step 10: Subarrays with K different integers

AI week 6: Ship an AI feature and test it
- 2 hours toward: A working endpoint, a before-and-after score table, and a resume line with one real number

– **Thu 18 Mar** Hard pass

Hard pass
- Step 10: Minimum window substring

AI week 6: Ship an AI feature and test it
- 2 hours toward: A working endpoint, a before-and-after score table, and a resume line with one real number

– **Fri 19 Mar** Hard pass

Hard pass
- Step 11: Maximum sum combination

AI week 6: Ship an AI feature and test it
- 2 hours toward: A working endpoint, a before-and-after score table, and a resume line with one real number

– **Sat 20 Mar** Hard pass

Hard pass
- Step 11: Find median from a data stream

AI week 6: Ship an AI feature and test it
- 2 hours toward: A working endpoint, a before-and-after score table, and a resume line with one real number

– **Sun 21 Mar** Hard pass

Hard pass

- Step 12: Candy

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

Semester exams

– WEEK 19 **22 Mar to 28 Mar 2027** SEMESTER EXAMS

College	Semester exams. Exam study comes first.
Minimum day	Revision 30 min, 2 problems, 1 LeetCode problem. Nothing else.

– WEEK 20 **29 Mar to 4 Apr 2027** SEMESTER EXAMS

College	Semester exams. Exam study comes first.
Minimum day	Revision 30 min, 2 problems, 1 LeetCode problem. Nothing else.

Finish and certify

– WEEK 21 **5 Apr to 11 Apr 2027**

College	Exams end around 7 Apr. Rest for two days, then restart at full pace.
DSA	Hard pass: 4 Hard problems, one a day, from Binary trees. 30 minutes alone before any hint.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 1100: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	Mixed viva: 5 questions a day across all five subjects.
Cyber	Lab week 15: SIEM. TryHackMe SOC Level 1 introductory rooms, or Splunk Free / Wazuh in a VM. Proof: Three saved searches that would catch an attack. One hour a day.
AWS	Second timed practice exam (aim for 80%+). Ask your faculty about an AWS Academy exam discount and book the exam for the week of 19 Apr.
AI	Week 7 of 12: OWASP Top 10 for LLM apps. Read all ten risks (prompt injection, sensitive information disclosure, system prompt leakage, excessive agency and the rest). Link each one to the web bug it resembles from cyber weeks 6 to 9. Proof: A one-page map: LLM risk, its web cousin, and the fix. Two hours a day.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.
Mock interviews	One mock interview. Note every question you could not answer.

DAY BY DAY

– **Mon 5 Apr** Semester exams

Semester exams
- Minimum day: revision, 2 problems, 1 LeetCode problem

– **Tue 6 Apr** Semester exams

Semester exams
- Minimum day: revision, 2 problems, 1 LeetCode problem

– **Wed 7 Apr** Semester exams

Semester exams
- Minimum day: revision, 2 problems, 1 LeetCode problem

– **Thu 8 Apr** Hard pass

Hard pass
- Step 13: Preorder, inorder and postorder in one traversal

AI week 7: OWASP Top 10 for LLM apps
- 2 hours toward: A one-page map: LLM risk, its web cousin, and the fix

– **Fri 9 Apr** Hard pass

Hard pass
- Step 13: Maximum path sum

AI week 7: OWASP Top 10 for LLM apps
- 2 hours toward: A one-page map: LLM risk, its web cousin, and the fix

– **Sat 10 Apr** Hard pass

Hard pass
- Step 13: Vertical order traversal

AI week 7: OWASP Top 10 for LLM apps
- 2 hours toward: A one-page map: LLM risk, its web cousin, and the fix

– **Sun 11 Apr** Hard pass

Hard pass
- Step 13: Print all nodes at distance K

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 22 **12 Apr to 18 Apr 2027**

DSA	Hard pass: 7 Hard problems, one a day, from Binary trees, Binary search trees. 30 minutes alone before any hint.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 1100: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	Mixed viva: 5 questions a day across all five subjects.
Cyber	Lab week 16: ATT&CK and incident response. MITRE ATT&CK navigator; the incident response lifecycle; map one machine's attack chain to ATT&CK techniques. Proof: An ATT&CK mapping table for that machine. One hour a day.
AWS	Light review of notes and wrong answers; one more practice exam if the last was under 80%.
AI	Week 8 of 12: Prompt injection, hands-on. Lakera's Gandalf game (as many levels as you can); the "Web LLM attacks" labs in PortSwigger Web Security Academy. Proof: A writeup of 2 labs, prompt by prompt. Two hours a day.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.
Mock interviews	One mock interview. Note every question you could not answer.

DAY BY DAY

– **Mon 12 Apr** Hard pass

Hard pass
- Step 13: Minimum time to burn a tree

AI week 8: Prompt injection, hands-on
- 2 hours toward: A writeup of 2 labs, prompt by prompt

– **Tue 13 Apr** Hard pass

Hard pass
- Step 13: Serialise and deserialise a binary tree

AI week 8: Prompt injection, hands-on
- 2 hours toward: A writeup of 2 labs, prompt by prompt

– **Wed 14 Apr** Hard pass

Hard pass
- Step 13: Morris inorder traversal

AI week 8: Prompt injection, hands-on
- 2 hours toward: A writeup of 2 labs, prompt by prompt

– **Thu 15 Apr** Hard pass

Hard pass

- Step 13: Morris preorder traversal

AI week 8: Prompt injection, hands-on

- 2 hours toward: A writeup of 2 labs, prompt by prompt

– **Fri 16 Apr** Hard pass

Hard pass

- Step 14: Delete a node

AI week 8: Prompt injection, hands-on

- 2 hours toward: A writeup of 2 labs, prompt by prompt

– **Sat 17 Apr** Hard pass

Hard pass

- Step 14: Recover a BST

AI week 8: Prompt injection, hands-on

- 2 hours toward: A writeup of 2 labs, prompt by prompt

– **Sun 18 Apr** Hard pass

Hard pass

- Step 14: Largest BST in a binary tree

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 23 **19 Apr to 25 Apr 2027** AWS EXAM, PLANNED

DSA	Hard pass: 7 Hard problems, one a day, from Graphs. 30 minutes alone before any hint.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 1200: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	Mixed viva: 5 questions a day across all five subjects.
Cyber	Lab week 17: Email and cloud. Phishing analysis: headers, SPF, DKIM, DMARC; AWS IAM least privilege, CloudTrail, GuardDuty (tie to TenantWall). Proof: TenantWall security notes: what is logged, who can do what. One hour a day.
AWS	Sit SAA-C03 this week. June is the latest.
AI	Week 9 of 12: Attack your own RAG app. Plant a poisoned note (indirect injection), try to leak the system prompt, misuse a tool. Then fix it: keep instructions and data apart, give tools the least power they need, confirm before actions. Proof: 3 findings, each with its fix and a retest. Two hours a day.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.
Mock interviews	One mock interview. Note every question you could not answer.

DAY BY DAY

– **Mon 19 Apr** Hard pass

Hard pass

- Step 15: Word ladder I

AI week 9: Attack your own RAG app

- 2 hours toward: 3 findings, each with its fix and a retest

– **Tue 20 Apr** Hard pass

Hard pass

- Step 15: Word ladder II

AI week 9: Attack your own RAG app

- 2 hours toward: 3 findings, each with its fix and a retest

– **Wed 21 Apr** Hard pass

Hard pass

- Step 15: Alien dictionary

AI week 9: Attack your own RAG app

- 2 hours toward: 3 findings, each with its fix and a retest

– **Thu 22 Apr** Hard pass

Hard pass

- Step 15: Print the shortest path

AI week 9: Attack your own RAG app

- 2 hours toward: 3 findings, each with its fix and a retest

– **Fri 23 Apr** Hard pass

Hard pass

- Step 15: Number of islands II

AI week 9: Attack your own RAG app

- 2 hours toward: 3 findings, each with its fix and a retest

– **Sat 24 Apr** Hard pass

Hard pass

- Step 15: Making a large island

AI week 9: Attack your own RAG app

- 2 hours toward: 3 findings, each with its fix and a retest

– **Sun 25 Apr** Hard pass

Hard pass

- Step 15: Swim in rising water

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– WEEK 24 **26 Apr to 2 May 2027**

DSA	Hard pass: 7 Hard problems, one a day, from Graphs, Dynamic programming. 30 minutes alone before any hint.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 1200: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	Mixed viva: 5 questions a day across all five subjects.
Cyber	Lab week 18: Play a CTF. Pick a beginner-friendly event on CTFtime; play solo or with classmates. Proof: Writeups for 2 challenges after the event ends. One hour a day.
AWS	Certified: add it to the resume and LinkedIn. If not yet sat, sit it now.
AI	Week 10 of 12: Reviewing AI-written code. Ask a model for 3 small features (login, file upload, a SQL query). Review them for security bugs, name each bug's CWE, and run Semgrep. Proof: Review notes: each bug, its CWE, and the fixed code. Two hours a day.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Internship	5 applications to good companies and one referral request. Stop when you have an offer.
Mock interviews	One mock interview. Note every question you could not answer.

DAY BY DAY

– **Mon 26 Apr** Hard pass

Hard pass

- Step 15: Bridges in a graph

AI week 10: Reviewing AI-written code

- 2 hours toward: Review notes: each bug, its CWE, and the fixed code

– **Tue 27 Apr** Hard pass

Hard pass

- Step 15: Articulation points

AI week 10: Reviewing AI-written code

- 2 hours toward: Review notes: each bug, its CWE, and the fixed code

– **Wed 28 Apr** Hard pass

Hard pass
- Step 15: Kosaraju's algorithm (strongly connected components)

AI week 10: Reviewing AI-written code
- 2 hours toward: Review notes: each bug, its CWE, and the fixed code

– **Thu 29 Apr** Hard pass

Hard pass
- Step 16: Cherry pickup II

AI week 10: Reviewing AI-written code
- 2 hours toward: Review notes: each bug, its CWE, and the fixed code

– **Fri 30 Apr** Hard pass

Hard pass
- Step 16: Shortest common supersequence

AI week 10: Reviewing AI-written code
- 2 hours toward: Review notes: each bug, its CWE, and the fixed code

– **Sat 1 May** Hard pass

Hard pass
- Step 16: Distinct subsequences

AI week 10: Reviewing AI-written code
- 2 hours toward: Review notes: each bug, its CWE, and the fixed code

– **Sun 2 May** Hard pass

Hard pass
- Step 16: Edit distance

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 25 **3 May to 9 May 2027**

DSA	Hard pass: 7 Hard problems, one a day, from Dynamic programming. 30 minutes alone before any hint.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 1300: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	Mixed viva: 5 questions a day across all five subjects.
Cyber	Lab week 19: Portfolio. Clean up a public security-notes repo: only writeups that platforms allow (your labs, retired machines, ended CTFs). Proof: Repo with 8 or more writeups and a README. One hour a day.
AI	Week 11 of 12: Machine learning for intrusion detection. Train a scikit-learn classifier for normal versus attack traffic on the NSL-KDD dataset, or for normal versus suspicious calls in TenantWall's CloudTrail logs. Measure precision, recall and false positives. Proof: A notebook with a confusion matrix, and a paragraph on what a false positive costs a security team. Two hours a day.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Mock interviews	One mock interview. Note every question you could not answer.

DAY BY DAY

– **Mon 3 May** Hard pass

Hard pass
- Step 16: Wildcard matching

AI week 11: Machine learning for intrusion detection
- 2 hours toward: A notebook with a confusion matrix, and a paragraph on what a false positive costs a security team

– **Tue 4 May** Hard pass

Hard pass

- Step 16: Best time to buy and sell stock III

AI week 11: Machine learning for intrusion detection

- 2 hours toward: A notebook with a confusion matrix, and a paragraph on what a false positive costs a security team

– **Wed 5 May** Hard pass

Hard pass

- Step 16: Best time to buy and sell stock IV

AI week 11: Machine learning for intrusion detection

- 2 hours toward: A notebook with a confusion matrix, and a paragraph on what a false positive costs a security team

– **Thu 6 May** Hard pass

Hard pass

- Step 16: LIS using binary search

AI week 11: Machine learning for intrusion detection

- 2 hours toward: A notebook with a confusion matrix, and a paragraph on what a false positive costs a security team

– **Fri 7 May** Hard pass

Hard pass

- Step 16: Matrix chain multiplication

AI week 11: Machine learning for intrusion detection

- 2 hours toward: A notebook with a confusion matrix, and a paragraph on what a false positive costs a security team

– **Sat 8 May** Hard pass

Hard pass

- Step 16: Minimum cost to cut a stick

AI week 11: Machine learning for intrusion detection

- 2 hours toward: A notebook with a confusion matrix, and a paragraph on what a false positive costs a security team

– **Sun 9 May** Hard pass

Hard pass

- Step 16: Burst balloons

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 26 **10 May to 16 May 2027**

DSA	Hard pass: 7 Hard problems, one a day, from Dynamic programming, Tries, Strings (advanced). 30 minutes alone before any hint.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 1300: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	Mixed viva: 5 questions a day across all five subjects.
Cyber	Lab week 20: Review and next target. Revisit weak areas; pick a certification target (for example CompTIA Security+ or eJPT) and draft a 3-month plan. Proof: A one-page plan for April to June. One hour a day.
AI	Week 12 of 12: Capstone polish and interview prep. Pick one project (the secured "Ask my notes" app or the intrusion-detection notebook): clean code, README, architecture diagram, a 2-minute demo. Answer 15 interview questions out loud (LLMs, embeddings, RAG, evaluation, prompt injection) and update your resume. Proof: One portfolio-ready repo, an updated resume, and your answers recorded. Two hours a day.
Aptitude	15 minutes a day: quant, logical and verbal, in rotation.
Mock interviews	One mock interview. Note every question you could not answer.

DAY BY DAY

– **Mon 10 May** Hard pass

Hard pass
- Step 16: Evaluate a boolean expression to true

AI week 12: Capstone polish and interview prep
- 2 hours toward: One portfolio-ready repo, an updated resume, and your answers recorded

– **Tue 11 May** Hard pass

Hard pass
- Step 16: Palindrome partitioning II

AI week 12: Capstone polish and interview prep
- 2 hours toward: One portfolio-ready repo, an updated resume, and your answers recorded

– **Wed 12 May** Hard pass

Hard pass
- Step 16: Maximum rectangle area with all 1s

AI week 12: Capstone polish and interview prep
- 2 hours toward: One portfolio-ready repo, an updated resume, and your answers recorded

– **Thu 13 May** Hard pass

Hard pass
- Step 17: Maximum XOR of two numbers in an array

AI week 12: Capstone polish and interview prep
- 2 hours toward: One portfolio-ready repo, an updated resume, and your answers recorded

– **Fri 14 May** Hard pass

Hard pass
- Step 17: Maximum XOR with an element from an array

AI week 12: Capstone polish and interview prep
- 2 hours toward: One portfolio-ready repo, an updated resume, and your answers recorded

– **Sat 15 May** Hard pass

Hard pass
- Step 18: Shortest palindrome

AI week 12: Capstone polish and interview prep
- 2 hours toward: One portfolio-ready repo, an updated resume, and your answers recorded

– **Sun 16 May** Hard pass

Hard pass
- Step 18: Rabin-Karp algorithm

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 27 **17 May to 23 May 2027** HARD PASS DONE

DSA	Hard pass: 3 Hard problems, one a day, from Strings (advanced). 30 minutes alone before any hint. The Hard pass ends this week.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 1400: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	Mixed viva: 5 questions a day across all five subjects.
Cyber	Binary taster 1 of 3: x86-64 basics (registers, the stack, calling conventions) in GDB with pwndbg. One hour a day.
Internship	With an offer, the internship starts around now. It replaces the capstone and most of the sprint; keep 1 hour of DSA revision a day.
Mock interviews	One mock interview. Note every question you could not answer.
Capstone	AI-assisted intrusion detection on TenantWall's CloudTrail logs: dataset, features, baseline model, evaluation.

DAY BY DAY

– **Mon 17 May** Hard pass

Hard pass
- Step 18: Z-function

– **Tue 18 May** Hard pass

Hard pass
- Step 18: KMP algorithm and the LPS array

– **Wed 19 May** Hard pass

Hard pass
- Step 18: Longest happy prefix

– **Thu 20 May** Revision

Revision
- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

– **Fri 21 May** Revision

Revision
- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

– **Sat 22 May** Revision

Revision
- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

– **Sun 23 May** Revision

Revision
- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

– WEEK 28 **24 May to 30 May 2027**

DSA	Two revision problems a day from the weakest topics in your error log.
LeetCode	One problem a day. Sunday contest, then upsolve one.
CP-31	Rating 1400: 2 problems a day. A Codeforces Div 3 or Div 4 round at the weekend.
Core CS	Mixed viva: 5 questions a day across all five subjects.
Cyber	Binary taster 2 of 3: simple crackmes in Ghidra; three picoCTF reverse-engineering challenges. One hour a day.
Mock interviews	One mock interview. Note every question you could not answer.
Capstone	Alerting demo, README, architecture diagram, a 2-minute demo video, and the resume line.

DAY BY DAY

– **Mon 24 May** Revision

Revision
- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

– **Tue 25 May** Revision

Revision
- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

– **Wed 26 May** Revision

Revision
- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

– **Thu 27 May** Revision

Revision

- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

– **Fri 28 May** Revision

Revision

- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

– **Sat 29 May** Revision

Revision

- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

– **Sun 30 May** Revision

Revision

- Redo problems marked H or F; one timed set of 3 problems in 90 minutes

Striver problems are in sheet order at 5 to 6 a day. If you fall behind, keep the order and carry on; do not skip ahead. AWS runs for 30 minutes and cyber for 30 minutes every day.

Placement sprint

– WEEK 29 **31 May to 6 Jun 2027**

DSA	Revise Striver, Steps 1 to 5: basics, sorting, arrays, binary search, strings. Re-solve from memory, about 15 problems a day.
Timed test	One full online-test simulation: 3 problems in 90 minutes, plus 20 aptitude questions.
Mock interviews	One full mock this week. Write down every question you could not answer.
Core CS	Answers out loud: OOP.
Profile	Resume final draft; get two people to review it.
CP and CSES	Optional, 1 hour: CP-31 from rating 1500, and the CSES section that matches this week's revision.
Stretch	Only if everything above is done: 30 minutes of probability for HFT.

– WEEK 30 **7 Jun to 13 Jun 2027**

DSA	Revise Striver, Steps 6 to 9: linked list, recursion, bits, stacks and queues. Re-solve from memory, about 15 problems a day.
Timed test	One full online-test simulation: 3 problems in 90 minutes, plus 20 aptitude questions.
Mock interviews	One full mock this week. Write down every question you could not answer.
Core CS	Answers out loud: Operating systems.
Profile	GitHub: pin 4 projects, each with README and diagram.
CP and CSES	Optional, 1 hour: CP-31 from rating 1500, and the CSES section that matches this week's revision.
Stretch	Only if everything above is done: 30 minutes of probability for HFT.

– WEEK 31 **14 Jun to 20 Jun 2027**

DSA	Revise Striver, Steps 10 to 13: sliding window, heaps, greedy, binary trees. Re-solve from memory, about 15 problems a day.
Timed test	One full online-test simulation: 3 problems in 90 minutes, plus 20 aptitude questions.
Mock interviews	One full mock this week. Write down every question you could not answer.
Core CS	Answers out loud: DBMS and SQL.
Profile	LinkedIn: post about TenantWall.
CP and CSES	Optional, 1 hour: CP-31 from rating 1500, and the CSES section that matches this week's revision.
Stretch	Only if everything above is done: 30 minutes of probability for HFT.

– WEEK 32 21 Jun to 27 Jun 2027

DSA	Revise Striver, Steps 14 and 15: BST and graphs. Re-solve from memory, about 15 problems a day.
Timed test	One full online-test simulation: 3 problems in 90 minutes, plus 20 aptitude questions.
Mock interviews	One full mock this week. Write down every question you could not answer.
Core CS	Answers out loud: Networks.
Profile	List 40 target companies; note which hire off-campus.
CP and CSES	Optional, 1 hour: CP-31 from rating 1500, and the CSES section that matches this week's revision.
Stretch	Only if everything above is done: 30 minutes of probability for HFT.

– WEEK 33 28 Jun to 4 Jul 2027

DSA	Revise Striver, Step 16: dynamic programming. Re-solve from memory, about 15 problems a day.
Timed test	One full online-test simulation: 3 problems in 90 minutes, plus 20 aptitude questions.
Mock interviews	One full mock this week. Write down every question you could not answer.
Core CS	Answers out loud: System design.
Profile	Ask alumni for referrals; apply to the first 10.
CP and CSES	Optional, 1 hour: CP-31 from rating 1500, and the CSES section that matches this week's revision.
Stretch	Only if everything above is done: 30 minutes of probability for HFT.

– WEEK 34 5 Jul to 11 Jul 2027

DSA	Revise Striver, Steps 17 and 18, then redo every problem marked F. Re-solve from memory, about 15 problems a day.
Timed test	One full online-test simulation: 3 problems in 90 minutes, plus 20 aptitude questions.
Mock interviews	One full mock this week. Write down every question you could not answer.
Core CS	Answers out loud: All five, mixed viva.
Profile	Company-specific prep for the first drives.
CP and CSES	Optional, 1 hour: CP-31 from rating 1500, and the CSES section that matches this week's revision.
Stretch	Only if everything above is done: 30 minutes of probability for HFT.

– WEEK 35 12 Jul to 18 Jul 2027

DSA	Revise Striver, Mixed timed sets: 3 problems in 90 minutes, daily. Re-solve from memory, about 15 problems a day.
Timed test	One full online-test simulation: 3 problems in 90 minutes, plus 20 aptitude questions.
Mock interviews	One full mock this week. Write down every question you could not answer.
Core CS	Answers out loud: All five, mixed viva.
Profile	Apply to 10 more; practise the project walk-through.
CP and CSES	Optional, 1 hour: CP-31 from rating 1500, and the CSES section that matches this week's revision.
Stretch	Only if everything above is done: 30 minutes of probability for HFT.

– WEEK 36 19 Jul to 25 Jul 2027

DSA	Revise Striver, Mixed timed sets and company-tagged problems. Re-solve from memory, about 15 problems a day.
Timed test	One full online-test simulation: 3 problems in 90 minutes, plus 20 aptitude questions.

Mock interviews	One full mock this week. Write down every question you could not answer.
Core CS	Answers out loud: HR answers and project walk-throughs.
Profile	Apply to 10 more; rest properly.
CP and CSES	Optional, 1 hour: CP-31 from rating 1500, and the CSES section that matches this week's revision.
Stretch	Only if everything above is done: 30 minutes of probability for HFT.

— WEEK 37 **26 Jul to 1 Aug 2027** PLACEMENTS OPEN ON 1 AUG

DSA	Revise Striver, Mixed timed sets; a light final week before the season. Re-solve from memory, about 15 problems a day.
Timed test	One full online-test simulation: 3 problems in 90 minutes, plus 20 aptitude questions.
Mock interviews	One full mock this week. Write down every question you could not answer.
Core CS	Answers out loud: HR answers and project walk-throughs.
Profile	Sleep, light revision, documents ready.
CP and CSES	Optional, 1 hour: CP-31 from rating 1500, and the CSES section that matches this week's revision.
Stretch	Only if everything above is done: 30 minutes of probability for HFT.

August 2027 to June 2028, by month

MONTH	PHASE	WHAT HAPPENS
Aug 2027	Final placements	First drives. Sit every test that fits. After each one, write down what was asked and fix that gap the same week. 10 off-campus applications and 5 referral requests a week. One contest and 10 problems a week to stay sharp.
Sep 2027	Final placements	Peak season. Keep the weekly rhythm: Monday shortlist, Tuesday apply, Wednesday referrals, Friday follow-ups. One mock interview every two weeks. Re-read the error log before every test.
Oct 2027	Final placements	If no offer yet: widen to cloud, DevSecOps and security roles, where TenantWall and the AWS certification carry more weight. Ask each interviewer for feedback.
Nov 2027	Final placements	Continue drives and off-campus. Start the final-year project proposal: extend TenantWall or the intrusion-detection work.
Dec 2027	Final placements	Season winds down. Review in writing: offer in hand, or the list of what interviews exposed and a 6-week plan to fix it.
Jan 2028	Final year	Final-year project build begins. If no offer: off-campus hiring continues, with 10 applications a week.
Feb 2028	Final year	Project build. Start a second certification: AWS Security Specialty or Security+.
Mar 2028	Final year	Project evaluation and results. Draft a second paper if the results are worth it.
Apr 2028	Final year	Sit the second certification. Project report.
May 2028	Final year	Final exams and project viva. With an offer: read up on your team's stack and system design.
Jun 2028	Final year	Graduate. First job starts in July.

HOURS BY BLOCK

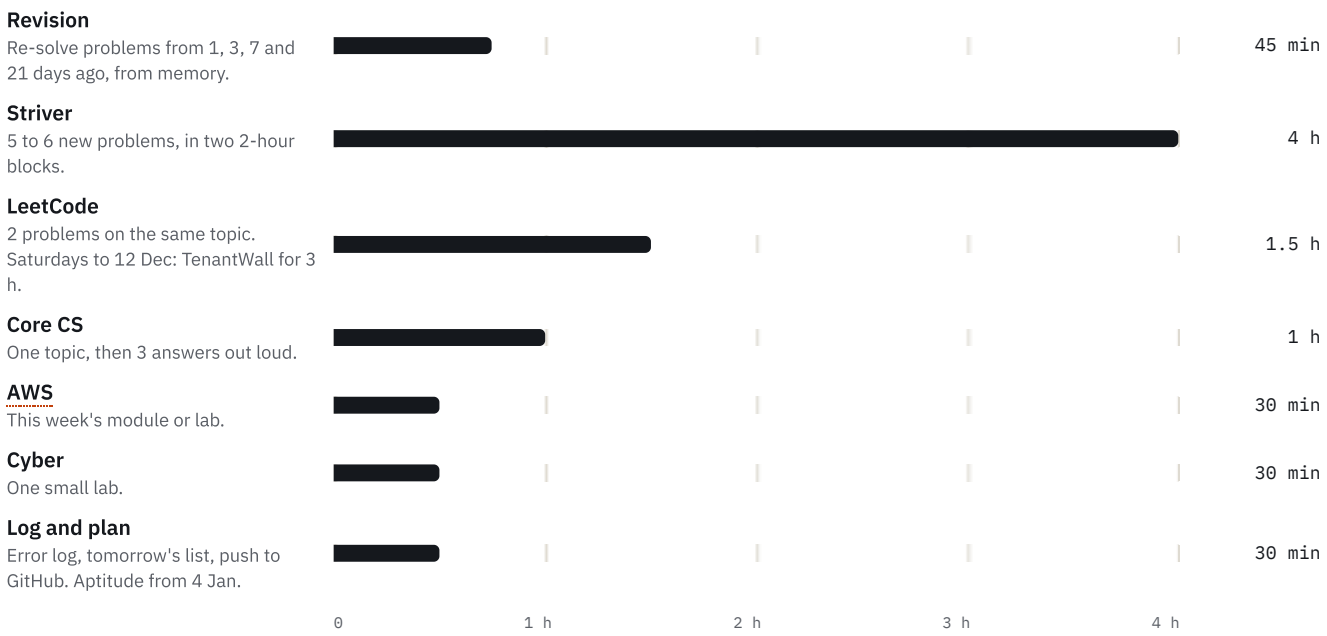
The day ↑ Index

How a day is split in each phase. Bars share one scale, so you can compare day types.

DSA heavy | Build | College day | Minimum day

DSA heavy · 18 Nov to 7 Feb, Monday to Saturday

Total 8 h 45 min



Sunday: the weekly contest, the written review, 2 hours of cyber, 4 hours on the backend project (from 20 Dec), then rest.

CHECKPOINTS

Milestones ↑ Index

Tick each one when it is true. "Latest" is how far a date can slip and still be on time for placements.

MILESTONE	PLAN	LATEST	STATUS
☐ Gate test passed, 4 of 5	17 Nov 2026		Due in 4 days
☐ Progress exported and the offline copy set up, before Claude Pro ends	18 Nov 2026		Due in 5 days
☐ TenantWall changes sent to the professor	12 Dec 2026		Ahead
☐ Checkpoint: Striver Step 6 reached	12 Dec 2026		Ahead
☐ Readiness check 1 filled in	20 Dec 2026		Ahead
☐ AWS course and its practice exam done	17 Jan 2027		Ahead
☐ Striver core done: all 359 Easy and Medium problems	30 Jan 2027	28 Feb 2027	Ahead
☐ Backend project deployed, with a README and diagram	31 Jan 2027	28 Feb 2027	Ahead
☐ Readiness check 2 filled in	15 Mar 2027		Ahead
☐ TenantWall paper submitted, before semester exams	21 Mar 2027		Ahead
☐ AWS Solutions Architect Associate passed	23 Apr 2027	30 Jun 2027	Ahead
☐ Bonus: a summer internship offer from a good company	30 Apr 2027		Ahead

☐	Hard pass done: 84 Hard problems attempted	19 May 2027	30 Jun 2027	Ahead
☐	AI projects on GitHub: RAG app, AI feature, security writeups, capstone	30 May 2027	30 Jun 2027	Ahead
☐	Placement-ready: Codeforces 1200 to 1400 and passing most mock interviews	31 Jul 2027		Ahead
☐	An offer in hand	31 Dec 2027		Ahead
☐	Graduated, with the paper published	30 Jun 2028		Ahead

Ticks save to this browser.

FINAL PLACEMENTS, AUGUST TO DECEMBER 2027

Targets ↑ Index

Three lanes. If a coding round goes badly at one company, the cloud and security lanes are still open. A summer 2027 internship at a good company is the bonus route: it can turn into a pre-placement offer.

ROLE	WHERE	WHAT THEY TEST	WITH THE PLAN DONE
SDE (main)	Product companies, startups, fintech	Coding tests, DSA, OOP, OS, DBMS, basic system design	Strong
Cloud or DevSecOps engineer	Cloud consultancies, platform teams	AWS, Linux, networking, scripting	Strong
Security engineer (AppSec or cloud)	Security product companies, banks, consulting cyber teams	OWASP, threat modelling, cloud IAM, code review	Strong
AI engineer (junior)	Startups building on LLMs	Python, RAG, evaluation, APIs	Medium
SDE at an HFT firm	Trading firms	Hard competitive programming, C++ depth, OS, networks, probability	Stretch

What you walk in with

- **DSA:** all 443 Striver problems, 84 Hard attempted, 250 or more LeetCode, CP-31 from 800 to 1400, a Codeforces rating
- **Core CS:** OOP, OS, DBMS, networks and system design, with 60 answers practised out loud
- **Cloud:** AWS Solutions Architect Associate
- **Security:** 20 weekly proofs of work, 8 or more writeups, a CTF, a STRIDE threat model
- **AI:** a RAG app, an AI feature in your backend project, 4 AI-security writeups, an intrusion-detection capstone
- **Backend:** a deployed REST API with a database, login and tests
- **Research:** the TenantWall paper, with audit and alerting added

What to expect, by how much you do

YOU DO	LIKELY RESULT
About 90%	Several offers: SDE at a product company, or a cloud or security role. HFT interviews are possible.
60 to 70%	SDE at a mid-size company, or a cloud or security role through your projects. Certification done.
Under 40%	DSA stays incomplete and coding tests become the wall. Cloud and security become the only path.

No plan guarantees a job. The market and which companies visit campus matter too, so apply off-campus as well.

THE PACE YOU CHOSE

The intensive track: 12 weeks to finish Striver core ↑ Index

The whole plan on this page now follows this track. For 12 weeks, DSA gets four hours a day and everything that does not help you pass a coding test, a Core CS round or a resume screen runs at low power.

WEEK	STRIVER, 5 TO 6 NEW PROBLEMS A DAY	ALSO
1 · 16 Nov	C++, STL; Step 1 Learn the basics	OOP; TenantWall
2 · 23 Nov	Step 1 Learn the basics, Step 2 Sorting techniques, Step 3 Arrays	OOP; TenantWall
3 · 30 Nov	Step 3 Arrays, Step 4 Binary search, Step 5 Strings (basic and medium)	Operating systems; TenantWall
4 · 7 Dec	Step 5 Strings (basic and medium), Step 6 Linked list, Step 7 Recursion and backtracking	Operating systems; TenantWall
5 · 14 Dec	Step 7 Recursion and backtracking, Step 8 Bit manipulation, Step 9 Stacks and queues	Operating systems
6 · 21 Dec	Step 9 Stacks and queues, Step 10 Sliding window and two pointers, Step 11 Heaps	Operating systems; TenantWall; Backend project; Resume version 1
7 · 28 Dec	Step 11 Heaps, Step 12 Greedy algorithms, Step 13 Binary trees	DBMS and SQL; Backend project; applications
8 · 4 Jan	Step 13 Binary trees, Step 14 Binary search trees	DBMS and SQL; Backend project; applications
9 · 11 Jan	Step 14 Binary search trees, Step 15 Graphs	DBMS and SQL; TenantWall; Backend project; applications
10 · 18 Jan	Step 15 Graphs, Step 16 Dynamic programming	DBMS and SQL; Backend project; applications; Mock interviews
11 · 25 Jan	Step 16 Dynamic programming, Step 17 Tries, Step 18 Strings (advanced)	Computer networks; Backend project; applications
12 · 1 Feb	Redo every H and F; mixed timed sets	Computer networks; Resume version 2; applications; Mock interviews

Why this pace. Six problems in four hours is 40 minutes each, which is what a careful beginner needs. Graphs and dynamic programming get 5 a day and are still the likeliest place to slip, so the week of 1 Feb is kept free to catch up.

The first check. On 12 December you should be at Step 6. If you are at Step 4 or lower, the pace is wrong for you: tell your coach, and move the end of Striver core to the end of February instead of pushing harder.

The playbook's dates. The playbook was built on an older, faster schedule, so its Overdue card will show a large number. Ignore the count. Tick problems there in sheet order, using this page for which ones are due.

METHODS THAT COMPOUND

Learn faster ↑ Index

Hours matter less than what you do inside them. These are the habits that change how much sticks.

Recall, do not re-read

Close the notes and write what you remember, then check. Re-reading feels smooth and teaches little. Every revision block is a re-solve from a blank file.

Space it: 1, 3, 7, 21 days

Re-solve each problem a day later, then at 3, 7 and 21 days. If it takes under 10 minutes twice, retire it. If you fail, the clock restarts.

Learn patterns, not problems

After each problem, write one line: the trigger that should have told you the technique. "Sorted array plus a target: two pointers or binary search." Interviews test the trigger.

Keep an error log

One line per mistake: off-by-one, overflow, wrong base case, misread constraint. Read it before every test. Most people repeat the same five errors.

Struggle on purpose

The 30 minutes before a hint are where the learning happens. If you never get stuck, the problems are too easy; if you are always stuck, drop a level.

Explain it out loud

Say the approach, the complexity and one edge case aloud before coding. If you cannot explain it simply, you do not know it yet. This is also exactly what an interview is.

Read constraints first

n up to 10^5 means O(n log n). n up to 20 means bitmasks or backtracking. The constraints often name the algorithm before you have read the story.

Practise under the clock

From week 6, one timed set a week: 3 problems in 90 minutes, no pausing. Speed and nerves are skills, and they only train under time.

Mix topics once you know them

Learning a topic: do it in a block. Revising: shuffle topics, so you practise choosing the technique, which is the hard part in a test.

Two deep blocks beat ten shallow hours

Two 2-hour blocks with the phone in another room, hardest work first. Take a 10-minute break every 50 minutes. Put easy work (videos, AWS) late in the day.

Protect sleep and movement

Seven to eight hours of sleep is when memory consolidates. A 20-minute walk a day keeps you sharp. Cutting sleep to study loses more than it gains.

Review weekly, in writing

Every Sunday, 20 minutes: what was solved, what failed and why, what to change. Then plan the week. Without the review you repeat the week instead of improving it.

For theory: questions, not notes

Turn every Core CS topic into 5 questions and answer them from memory. NotebookLM can quiz you. Notes you never test yourself on are decoration.

Build to understand

A linked list you implemented, a tiny LRU cache, a rate limiter: one small build per topic makes the idea permanent and gives you something to talk about.

THE COMPLETE SYSTEM

Off-campus

↑ Index

Off-campus hiring is a numbers game played with a good profile. Expect roughly 100 applications for 5 to 10 tests and 2 to 4 interviews. Referrals change those odds more than anything else.

Where openings appear

- Company careers pages: check your 40-company list twice a week
- LinkedIn Jobs, with alerts for "SDE intern", "software engineer 2028", "security intern"
- Fresher programmes and hiring challenges run by large companies
- Coding contests and hackathons that end in interviews
- Startup boards such as Wellfound and Y Combinator's Work at a Startup
- Alumni and seniors: the fastest route, through a referral

Your profile, before the first application

- **Resume:** one page, PDF, no photo. Order: education, projects, skills, achievements
- **Each bullet:** what you built, with what, and a number. "Cut IAM policies from N per team to 2 using ABAC; 14 of 14 isolation tests pass."
- **GitHub:** 4 pinned repos, each with a README, a diagram and how to run it
- **LinkedIn:** a headline that says what you do, an About section, the projects, the AWS certification when you have it
- **Coding profiles:** LeetCode and Codeforces links, once the numbers are worth showing

The weekly rhythm

- **Monday:** 30 minutes scanning your sources; shortlist 5 to 10 roles
- **Tuesday:** apply, tailoring the top 3 resume bullets to each role
- **Wednesday:** 5 referral messages
- **Friday:** follow up on anything a week old, once
- **Always:** log every application: company, role, link, date, referral, status, next step
- Never pay anyone for a job, an interview or a "training bond". Real companies do not charge.

Referral message to an alumnus

```
Hi [Name], I'm a third-year CSE (Cybersecurity) student at RAIT,
graduating in 2028. I saw the [role] opening at [company] (job ID [ID]).

I've built TenantWall, a multi-tenant access-control system on AWS that
our team wrote up as a paper, and I've solved [N] DSA problems on
LeetCode.

Would you be willing to refer me? My resume is attached, and I can send
anything else you need. Thank you for your time.
```

Keep it under 100 words. Name the exact role and job ID. Send it to people one to four years ahead of you; they remember the search.

Message to a recruiter or engineer you do not know

```
Hi [Name], I'm a 2028 CSE graduate from RAIT focused on backend and
cloud security. I built [one-line project with a number] and hold [AWS
certification, once earned].

I'm interested in [role] at [company]. Is there an intern or new-
graduate opening I should apply to, or someone I should speak with?
Resume attached.
```

Expect about 1 reply in 10. That is normal. Follow up once after a week, then move on.

ROUND	WHAT IT IS	HOW TO PREPARE
Resume screen	Seconds of reading, often by keyword	Use the job post's own words for skills you really have. Numbers in every project bullet.
Online test	2 to 3 coding problems in 60 to 90 minutes, sometimes MCQs on CS, SQL and aptitude	Weekly timed sets. Read all problems first, solve the easiest fully, then go for partial marks. Practise aptitude 15 minutes a day in the last month.
Technical interview 1	1 or 2 DSA problems, talking as you go	Restate, give the brute force, improve it, code it, test it aloud. Ask about constraints. Silence loses more than a wrong first idea.
Technical interview 2	Projects in depth, Core CS, sometimes basic design	For each project: why this design, what broke, what you would change. Know every line. Prepare the TenantWall walk-through to 3 minutes.
Security roles	OWASP, a code-review exercise, threat modelling, your labs and CTFs	Explain one bug you found end to end. Threat-model TenantWall with STRIDE. Review code for the classic five: injection, broken access control, secrets, unsafe deserialization, missing validation.
HR or manager	Motivation, teamwork, a conflict, a failure	Five stories in situation, task, action, result form: a hard bug, a team disagreement, a deadline, a failure, something you taught yourself.

In the interview

- Think aloud from the first second; the interviewer is grading your reasoning
- Ask for the constraints and an example before writing anything
- State the complexity without being asked
- When stuck, say what you are considering and why it fails; hints are part of the process
- Test with a small case and an edge case before saying "done"
- End with one real question about the team's work

After every test or interview

- Write down every question the same day
- Solve the ones you missed within 48 hours
- Add the gap to that week's plan
- Rejections are data, not verdicts: most strong candidates collect dozens
- With an offer: get it in writing, check the role, location, pay and joining date, and ask the college about the NOC before accepting

DECIDED IN ADVANCE

Rules ↑ Index

These settle the hard calls before the bad day arrives.

Minimum day, never zero

When time is short, cut in this order

On a bad day: revision, 2 Striver problems, 1 LeetCode problem and 20 minutes of Core CS flashcards. A missed day must not become a missed week.

Think before you look

30 minutes alone before a hint. 60 before a solution. Then close it, code it from memory, and re-solve it the next day.

AI is a tutor, not an answer machine

Ask for hints, reviews and quizzes. Never on a first attempt, never in a gate test. In projects, read and understand every line it writes.

If you are below Step 6 on 12 Dec

Do not push harder. Move the end of Striver core to the end of February and start AI at 1 hour a day.

1. HFT preparation
2. CP-31
3. AI, down to 1 hour
4. Cyber, down to Sundays only

Never cut DSA, revision or Core CS.

Videos at the right moment

Concept videos before a new topic, at 2×. Solution videos only after your own attempt.

Limits

Stop at 9 hours. Take one half-day off every week.

If Striver is not finished on 7 Feb

CP-31 waits until it is, and AI runs at 1 hour.

MOSTLY FREE

Toolkit ↑ Index

Nothing in this plan needs a paid course.

FOR	USE	WHEN
DSA for interviews	Striver's A2Z sheet on takeUforward, in the playbook's order; LeetCode	From 16 Nov
Contest speed	The CP-31 sheet and Codeforces rounds	From 14 Dec
Harder problems	CSES problem set with the Competitive Programmer's Handbook	April to July
Core CS	Playbook Part 04; NotebookLM for flashcards and quizzes from your notes	Daily, 45 min
Cyber	TryHackMe, OverTheWire Bandit, PortSwigger Web Security Academy, Hack The Box, picoCTF	Daily, 1 h
Cyber videos	David Bombal for networking concepts; LiveOverflow for web and binary walkthroughs. Watch before the matching lab, not instead of it.	As each topic comes up
Binary taster	GDB with pwndbg, Ghidra, picoCTF reverse-engineering challenges; pwn.college if you want more	19 Apr to 9 May
AWS	AWS Academy Cloud Architecting, the free tier with a billing alarm, the SAA-C03 exam guide	Daily, 30 min
AI	Google AI Studio's free API or Ollama; Gandalf and PortSwigger's LLM labs for security	From 4 Jan
Gap check	roadmap.sh: tick the Backend, Cyber Security, AWS and AI Engineer maps for 30 minutes	Once a month
Probability (stretch)	Harvard Stat 110 with Blitzstein's book, Brainstellar, Zetamac	April to July
AI assistants	Gemini student plan every day; Claude's free plan for bigger one-off tasks	From 19 Nov
Claude Pro with Claude Code	Worth paying for in months when you build projects, not in months when you mostly study	January to May, as needed

STILL OPEN

Decisions that change the plan ↑ Index

- **What is the minimum attendance your college requires?** You plan to attend 1 to 2 days a week from 14 Dec. Check the rule for sitting exams and for placement eligibility, and stay just above it.
- **When is the TenantWall deadline, and where will it be submitted?** Ask the professor. If it is before mid-December, the four Saturdays move earlier.
- **What did the professor mean by "something explicit"?** Most likely explicit deny. Confirm before writing the section.
- **Which companies count as "good" for an NOC?** The college allows an internship at a good company. Ask the placement office for the criteria before you apply, so every application is one you could accept.

Daily problems, ticks and XP live in the DSA Placement Playbook. This page is the map.

Plan as of 13 Nov 2026 · intensive track